

Variant calling & Annotation

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2.10.2025

Institute of Tissue Medicine and Pathology – University of Bern
Clinical Genomics Lab - **INSELSPITAL**

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Part I

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Github Cancer Genomics Course



<https://github.com/supermax1234/CancerGenomicsCourse>

Comparing individual human genome sequences

**Reference
genome sequence**

[illegible]

**Your
genome sequence**

ATGTCGTTTTCGTAGTATACGCTAGGCGTCAACGCCCATCTGTGTCAGGTGGCCGACGAGACA
CTGTCCTGTATTCCTCCGCTCTTAATAGACACACAGGGGGCAATACAGCAGCAAGGACGCTTCGCGAC
CAACGCTCGTCAGCAAAACGAAAGAGCTTAAGCGTCGCCAATTCGCGATTCGCGGTGCGTCTGGGTG
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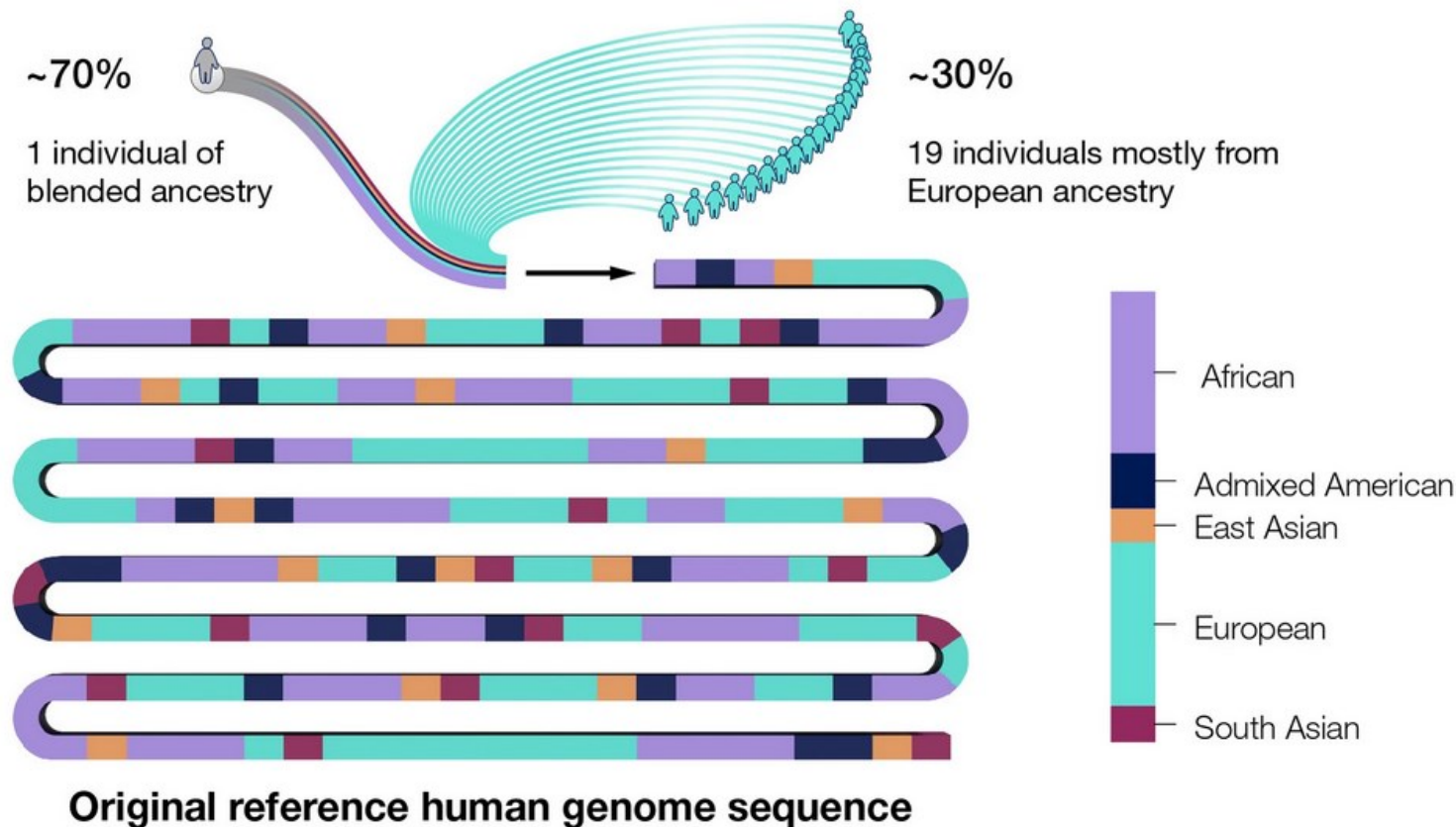
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TAGTGTGATTATCGGAGA

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TTAACCTGAGTGAGTTACTAACGGTATTCTCGGGATTA-- --GTATAGTGAGTATCGGAGA
CAAGCGCATCTCGCAGCAAGCTGTGTCAGCAACGAAAGGCTTAAGGCTGCCAATTCGCGATGT
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TGTGGTCTGCGGTGCAAAATGTCGGGCATACGTATTGTCTGCAGCTGTTTGTCTCAGCAAGTTCG
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TATGATAGCCCGGCCCTTATTAGTACAGTAGTGCTTGAATGGCATCTCTTTATTAAGTTTCTCTA
CAGCTAAACGATCAAGTGCATCTCCAGAGACACTGTCCGTGATTTCTCCGCTCTTAATAGCACAC
CGGGGCAATACGACGACAGCAAGCTGTGCGACAGCTCGTCGACGAAACGAAAGGATTTAAGGCT
CGCCAAATTCRCATCTCAGGGTGCGTGGGTGTTTGTGATCGGCTCAGGTACGCTAGTATGCGT
TCTCTTCCTTCAGGGGATGTGGCTGCGTGGTCAAAATGAGGCATACGATTGCTGCGAGTGT

The Reference Genome: A Standardized Map

The reference genome serves as a **universal map** for comparing genomic sequences. It is a **composite, haploid sequence** representing the **most common alleles** across a population. It's crucial to understand that it's **not a single individual's genome** but a standardized template that provides a common coordinate system.



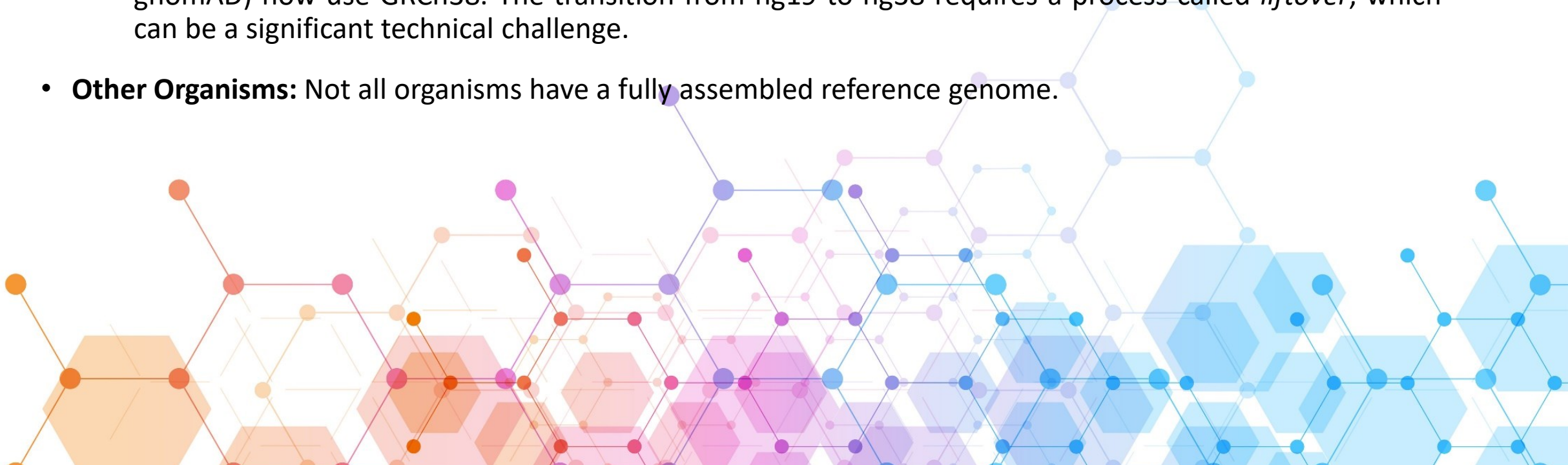
Detecting variants in a person's genome sequence usually requires an existing sequence for comparison — a *reference*.

A reference human genome sequence is an *established, high-quality* and *well-accepted sequence* of a human genome (i.e., one sequence for each of the 23 human chromosomes).

A reference human genome sequence is not an actual sequence from one individual but pieced together from multiple people. The important feature of a reference human genome sequence is that it depicts one assigned nucleotide for every position across the human genome. In this regard, a reference human genome sequence only represents the sequence of one copy of each chromosome, whereas a person's genome sequence contains the sequences of both copies of each chromosome (i.e., people are diploid).

The Reference Genome: A Standardized Map

- **The Human Genome Reference Sequence:**
 - **Current Standard (de-facto): hg19 / GRCh37** : These two versions are functionally the same and are still very widely used, especially in older datasets and clinical pipelines. Many established tools and databases (like dbSNP and 1000 Genomes) are built on this assembly.
 - **Newer Standard: GRCh38**: This is the current, more complete reference assembly. It includes improvements such as more accurate centromeric regions and the incorporation of *alternative loci* to better represent human genetic diversity. All major consortia (like the Genome Aggregation Database, gnomAD) now use GRCh38. The transition from hg19 to hg38 requires a process called *liftover*, which can be a significant technical challenge.
- **Other Organisms**: Not all organisms have a fully assembled reference genome.



Liftover

Lift over variants, positions, or intervals from one reference genome to another.

This web interface supports more input formats and requires fewer clicks than existing interfaces like [UCSC LiftOver](#) and [Ensembl Assembly Converter](#).

Under the hood, it uses the [UCSC liftOver command-line tool](#) for **intervals** and **positions**, while for **variants** it uses a new [bcftools plugin](#) which avoids many of the limitations of other liftover tools (particularly for indels) as described in [\[Genovese 2024\]](#).

To submit issues or feature requests, please use [broadinstitute/liftover/issues](#).

Input formats:

- Positions:**
chr8:141310715
chr8-141310715
chr8 141310715
- Variants:**
chr8:141310715 T>G
chr8-141310715-T-G
NM_001089.3(ABCA3):c.875A>T (p.Glu292Val)
- Intervals:**
chr8:141310715-141310720
chr8 141310715 141310720

The "chr" prefix is optional, and separators can be `:`, `-`, `>`, or space(s).

☒ hg19 => hg38

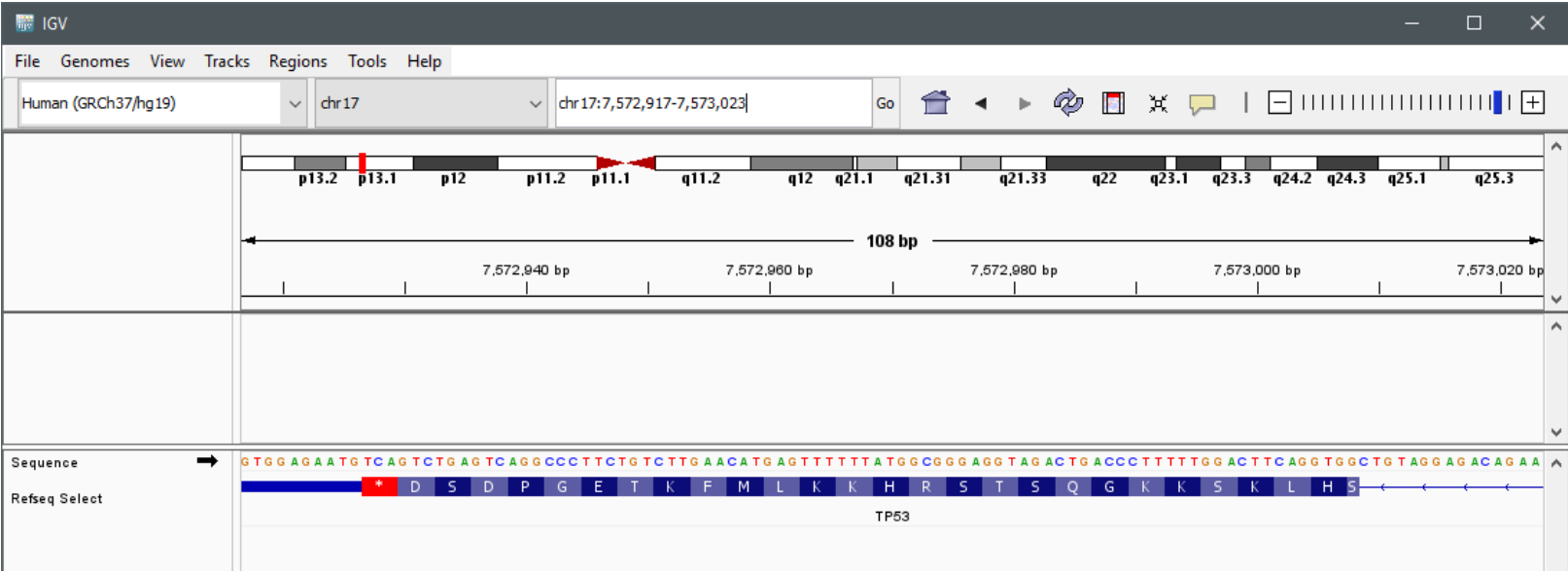
☐ hg38 => hg19

☐ hg38 => T2T

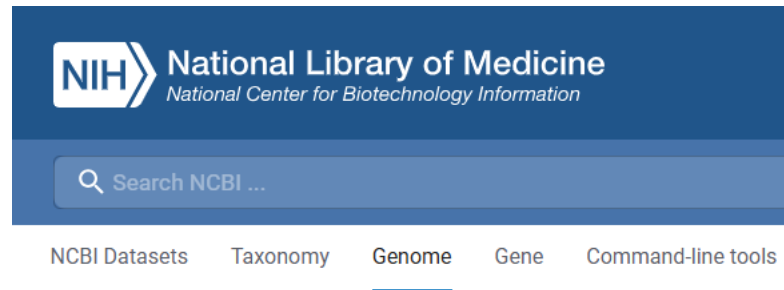
☐ T2T => hg38

Submit

<https://liftover.broadinstitute.org/#>



National Center for Biotechnology Information (NCBI)



Genome assembly GRCh37.p13

Download

datasets

API

FTP

See latest version: [GCF_000001405.40](#)

NCBI RefSeq assembly GCF_000001405.25 (replaced)

Submitted GenBank assembly GCA_000001405.14 (replaced)

Taxon [Homo sapiens](#) (human)

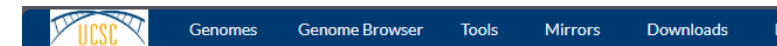
Synonym hg19

Assembly type haploid with alt loci

Submitter Genome Reference Consortium

Date Jun 28, 2013

UCSC Genome Browser



Sequence, Annotation, and Other Downloads

This page contains links to sequence and annotation downloads for the genome assemblies from the [FTP server](#). Data filtering is available in the [Table Browser](#) or via the command-line [utilities](#).

For access to the most recent assembly of each genome, see the [current genomes](#) directory. [GenArk](#) (Genome Archive) species data can be found [here](#). All data in the download directories. These data were contributed by many researchers, as listed on the [Genome Browser](#).

[Human](#)

[Mouse](#)

[SARS-CoV-2 \(COVID\)](#)

[Zebrafish](#)

[Fruit fly](#)

[Mammals](#) ▶

Human genomes

Jan. 2022 (T2T-CHM13 v2.0/hs1)

This assembly represents the [T2T-CHM13v2.0](#) genome. While it may be more recent than

This assembly is served entirely as a [track hub](#), meaning no MySQL files exist.

- [Fileserver \(bigBed, maf, fa, etc\) annotations](#)
- [Genome sequence files](#)
- [Track hub base directory](#)
- [LiftOver files](#)
- [Pairwise alignments](#) ▶

Dec. 2013 (GRCh38/hg38)

- [Genome sequence files and select annotations \(2bit, GTF, GC-content, etc\)](#) ▶
- [Sequence data by chromosome](#)
- [Annotations](#) ▶
- [SNP-masked fasta files](#) ▶
- [LiftOver files](#)
- [Pairwise alignments](#) ▶
- [Multiple alignments](#) ▶
- [Patches](#) ▶

<https://www.ncbi.nlm.nih.gov/assembly/37871>

<https://hgdownload.soe.ucsc.edu/downloads.html#human>

Alternative Site: Ensembl

[BLAST/BLAT](#) | [VEP](#) | [Tools](#) | [BioMart](#) | [Downloads](#) | [Help & Docs](#)

Search

All species ▼

for

Go

e.g. [BRCA2](#) or [human 5:62797383-63627669](#) or [rs699](#) or [coronary heart disease](#)

Browse a Genome

The Ensembl project produces genome databases for vertebrates and other eukaryotic species, and makes this information freely available online.

Available genomes



Human
GRCh37.p13

Want to use GRCh38?

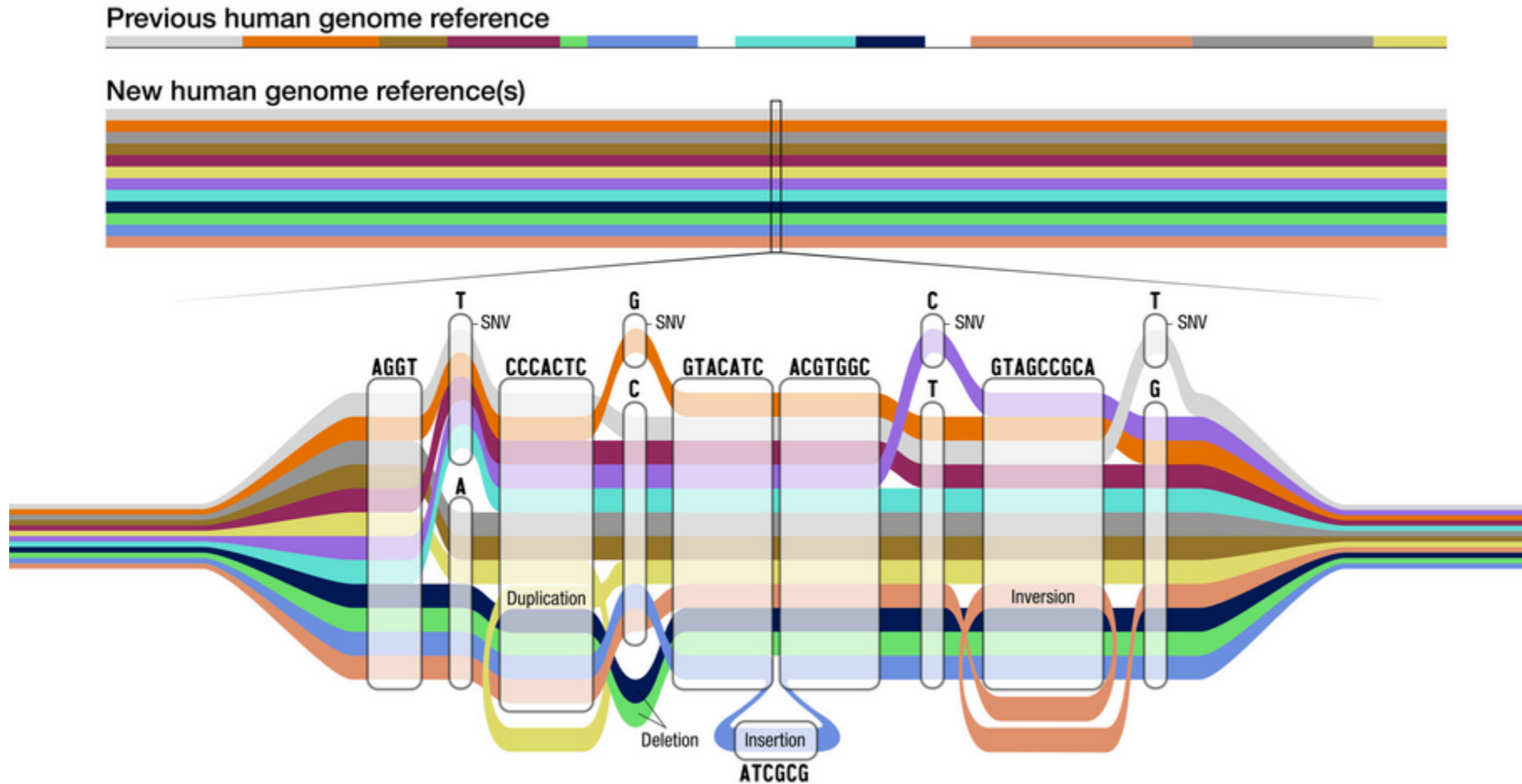


Our [main site](#) features the GRCh38 *Homo sapiens* assembly, with the latest gene models, variants, regulatory build and more!

[Learn more](#) about how to migrate your data to GRCh38

<https://grch37.ensembl.org/>

What Is a Pangenome Reference Sequence and How Is It Used?



Reference human genome sequences are invaluable for detecting genomic variants in each newly sequenced human genome. But the currently available reference human genome sequences do not accurately represent the genomic diversity of the human population, and that lack of **diversity** can introduce **biases** when analyzing some peoples' genome sequences.

To address this deficiency, researchers are working to generate a more complete set of reference human genome sequences that better reflect all of humanity. A "pangenome" is the collective genome sequences of multiple individuals that better represents the genomic diversity of the species. The human pangenome reference sequence will provide a better tool for comparing genome sequences from people all over the world in an effort to detect and characterize genomic variants more completely, including those with important roles in health and disease.

Human Pangenome Reference Consortium (HPRC)

[Sequencing Data](#)[Assemblies](#)[Annotations](#)[Alignments](#)[HPRC Website](#)[Help & Documentation](#)

Filters

[Clear All](#)

SRA

Biosample Accession (232) ▾

SAMPLE

Sample ID (232) ▾

SEQUENCING

Platform (3) ▾

Instrument Model (6) ▾

Library Strategy (3) ▾

Filetype (3) ▾

Basecaller (5) ▾

Basecaller Version (11) ▾

Sequencing Data

Table

Graph

Results 1 - 6048 of 6048

[Download TSV](#)

Group By ▾

Edit Columns ▾

	Filename ↑	Sample ID	Family ID	Population Abbreviation	Population Descriptor	Biosample Accession	Platform
Download	BN001_R1.trimmed.fastq.gz	HG00733	Unspecified	PUR	Puerto Rican in Puerto Rico	SAMN00006581	ILLUMINA
Download	BN001_R2.trimmed.fastq.gz	HG00733	Unspecified	PUR	Puerto Rican in Puerto Rico	SAMN00006581	ILLUMINA
Download	GM20503.deepconsensus.fastq.gz	NA20503	Unspecified	TSI	Toscani in Italia	SAMN41021630	PACBIO_SMRT
Download	GM20762.deepconsensus.fastq.gz	NA20762	Unspecified	TSI	Toscani in Italia	SAMN41021652	PACBIO_SMRT
Download	GM20806.deepconsensus.fastq.gz	NA20806	Unspecified	TSI	Toscani in Italia	SAMN41021648	PACBIO_SMRT
Download	GM20827.deepconsensus.fastq.gz	NA20827	Unspecified	TSI	Toscani in Italia	SAMN41021650	PACBIO_SMRT
Download	HG002.HiC_1_NovaSeq_1_S1_L001_R1_001.fastq.gz	HG002	Unspecified	Unspecified	Unspecified	SAMN03283347	ILLUMINA

<https://data.humanpangenome.org/>

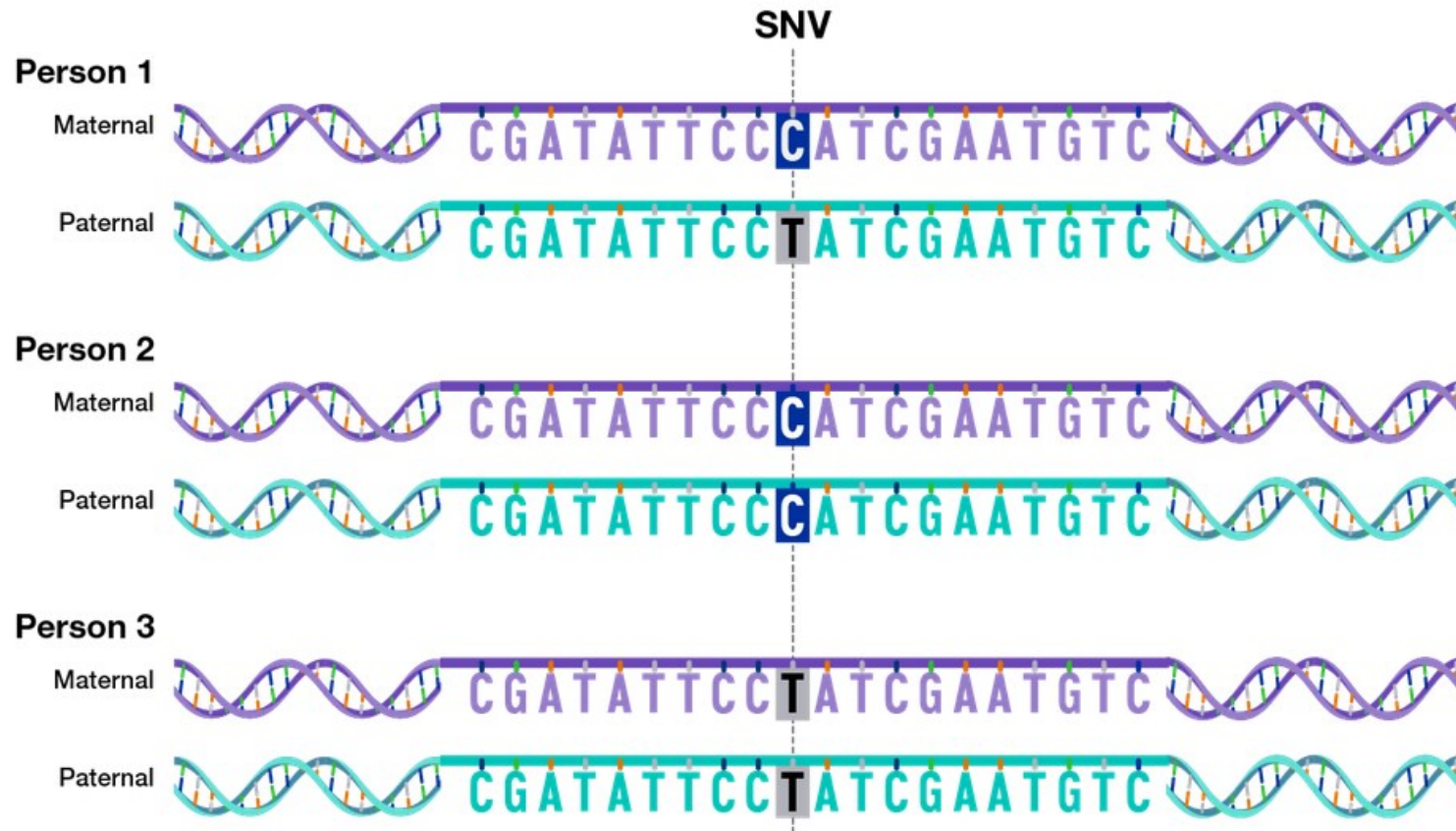
Why Variation Matters?

The vast **majority of the DNA letters** in peoples' genomes is **identical**, but a small fraction of those letters varies. This **genomic variation** accounts for some of the differences among people, including important aspects of their *health* and *susceptibility to diseases*.

- **Genomic variation** reflects the differences in a person's DNA compared to other peoples' DNA.
- There are multiple types of variants in human genomes, ranging from **small** differences to **large** differences.
- A very small subset of genomic variants contributes to human health and disease.
- Researchers create reference **human genome** sequences to help **detect genomic variants** in each sequenced human genome.
- On average, a person's genome sequence is **~99.6% identical** to a reference **human genome** sequence; that person's set of genomic variants accounts for the **~0.4% difference**.
- The *human pangenome* is a more comprehensive framework that aims to account for **genomic variation** across human populations, thereby reducing biases that can come with the use of a single reference **human genome** sequence.

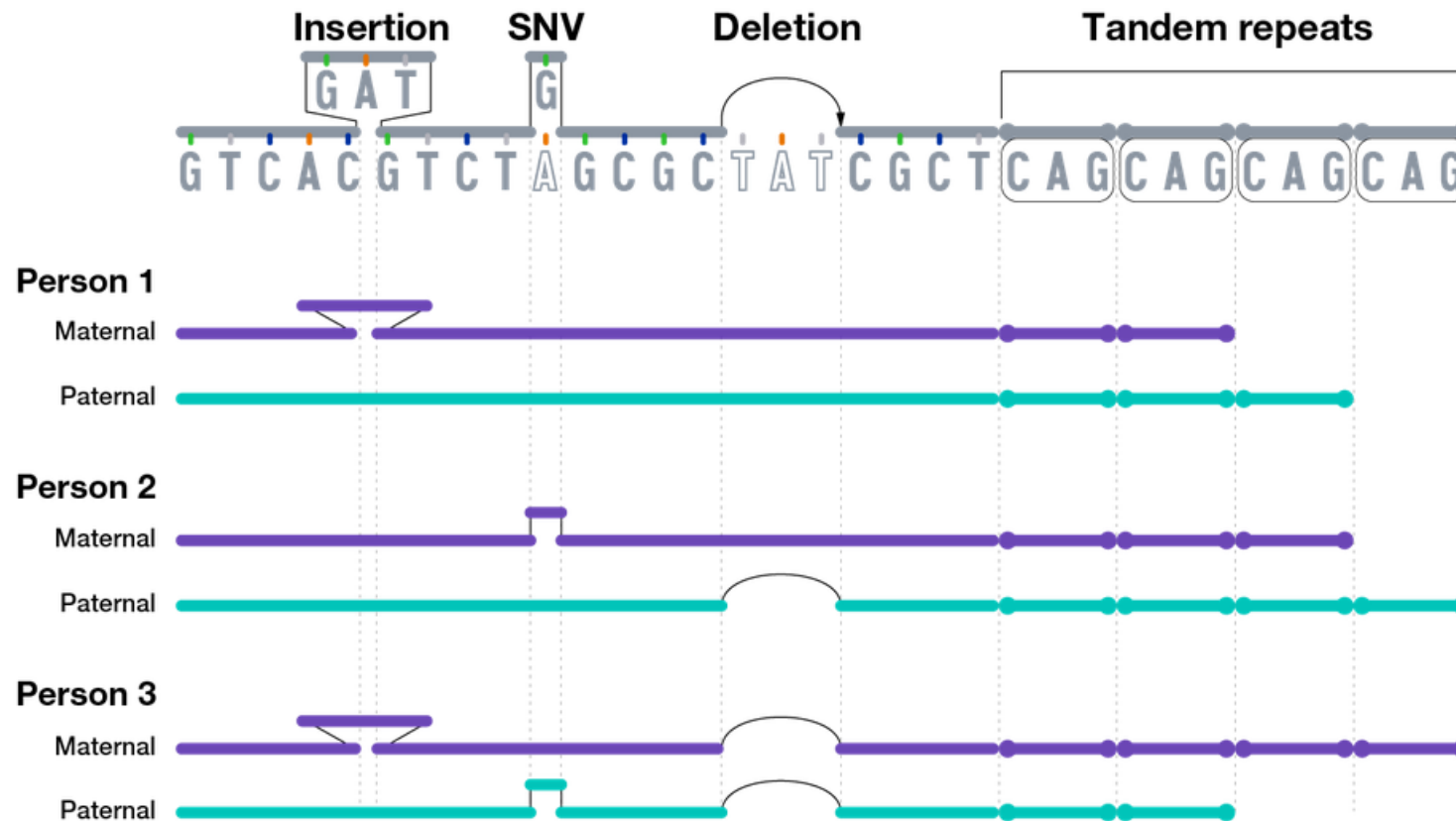


What Are the Different Types of Genomic Variants?



The smallest genomic variants are **single-nucleotide variants** (SNVs). Each SNV reflects a difference in a single nucleotide. For a given SNV, the DNA letter at that genomic position might be a C in one person but a T in another person. SNVs are the most common type of **genomic variation**. A subtype of SNVs is called a single-nucleotide polymorphism (SNP; pronounced "snip"). To be considered a SNP, a SNV must be present in **at least 1% of the human population**. As such, SNV is a more general term that includes both relatively common (such as SNPs) and rare single-nucleotide differences. For simplicity, we refer to all single-nucleotide differences as SNVs, regardless of their relative frequency.

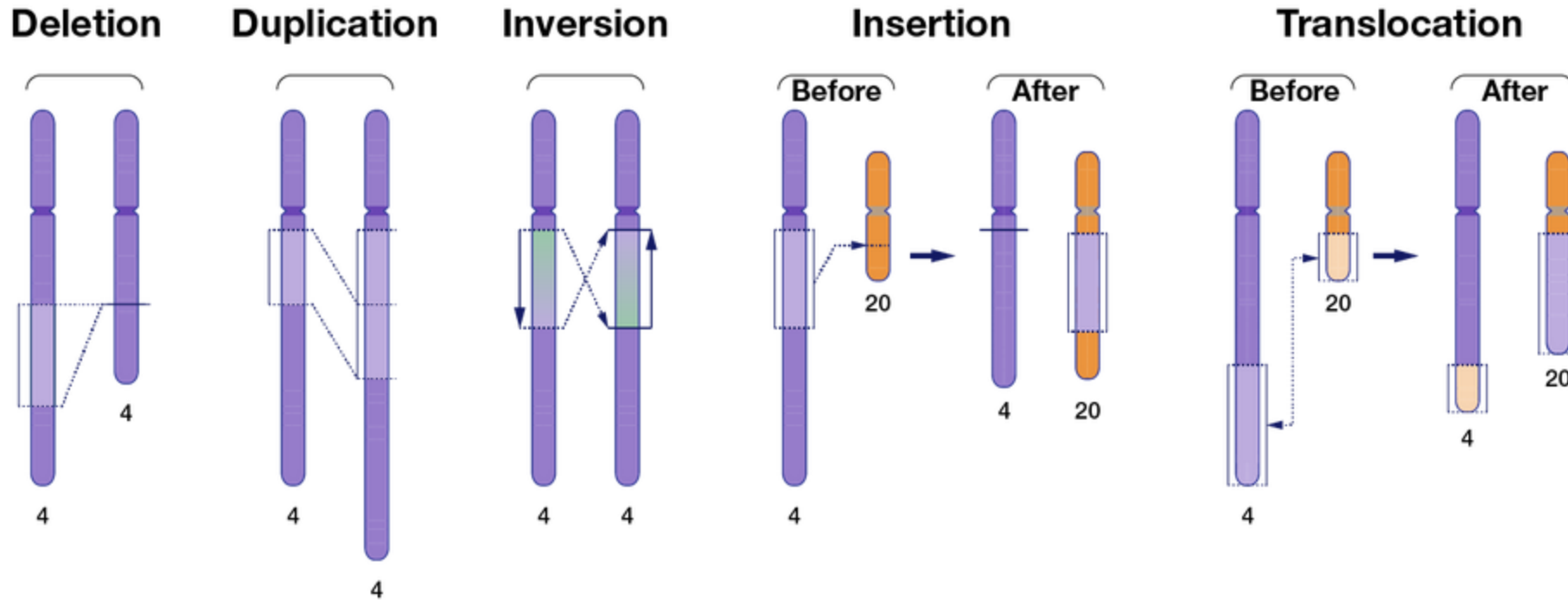
What Are the Different Types of Genomic Variants?



Another group of small genomic variants are **insertions** and **deletions** (*indels*). Insertion/deletion variants reflect extra or missing DNA nucleotides in the genome, respectively, and typically involve fewer than 50 nucleotides. Insertion/deletion variants are less frequent than SNVs but can sometimes have a larger impact on health and disease.

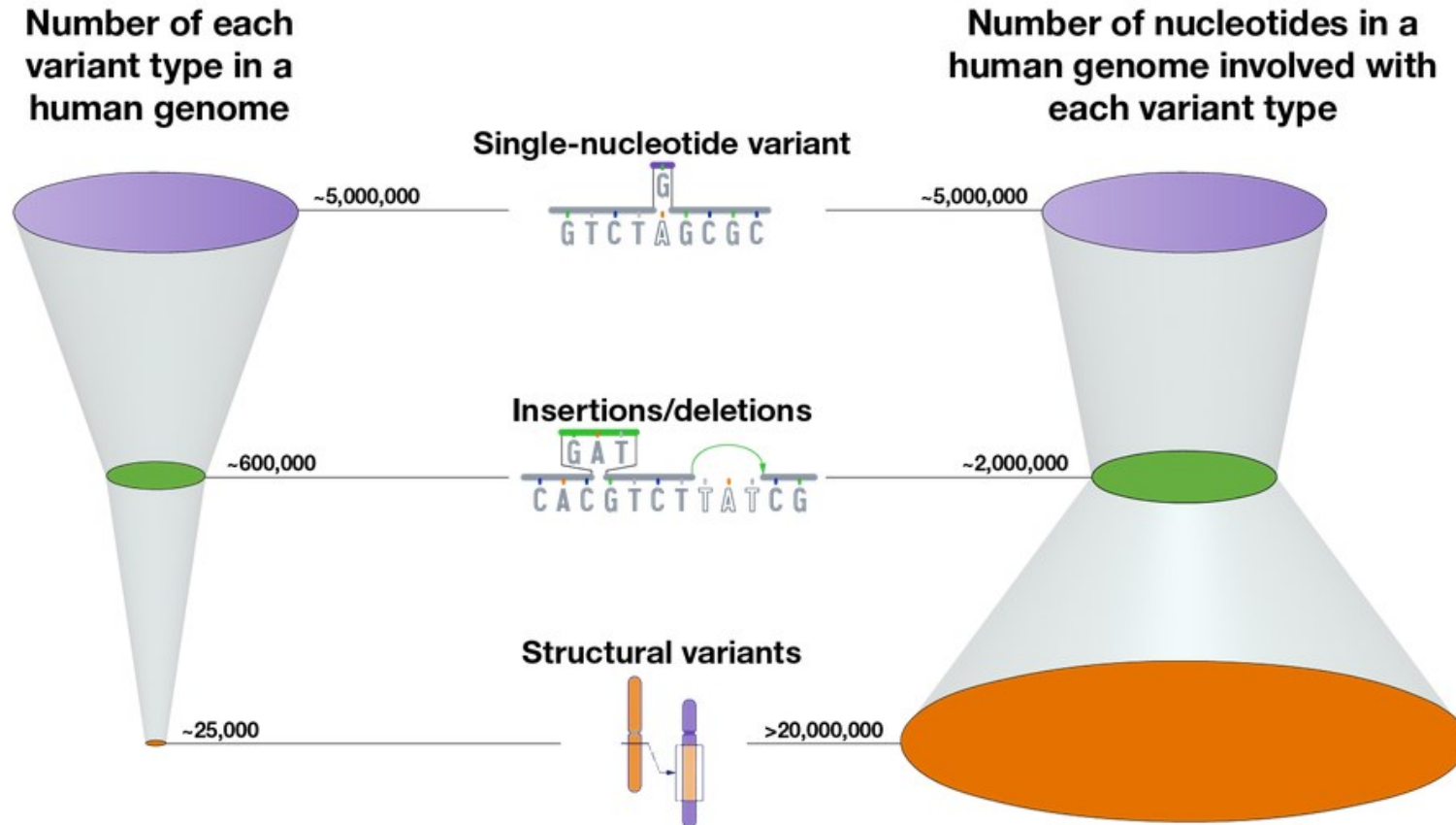
One of the most common types of insertion/deletion variants are tandem repeats (also known as **microsatellites**). Tandem repeats are short stretches of nucleotides that are repeated multiple times and are highly variable among people. Different **chromosomes** can vary in the number of times such short nucleotide stretches are repeated, ranging from a few times to hundreds of times.

What Are the Different Types of Genomic Variants?



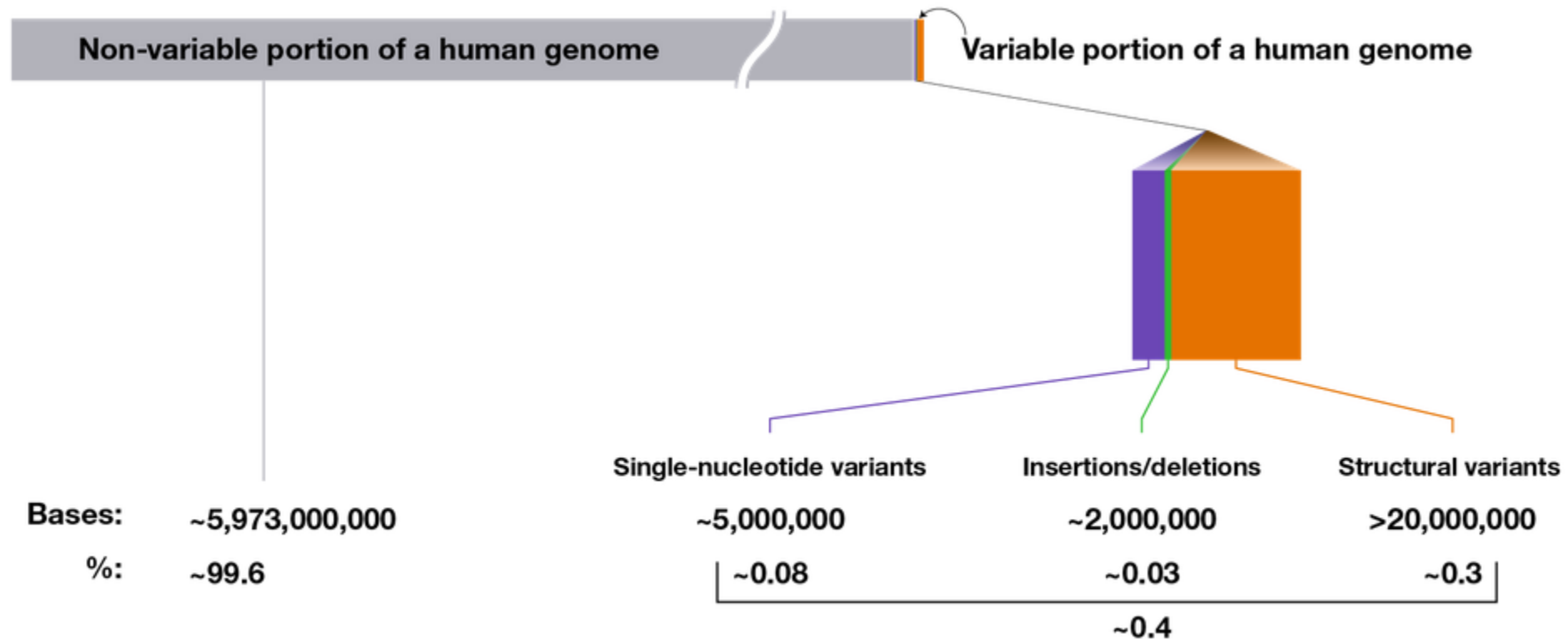
Genomic variation also extends beyond small stretches of nucleotides to larger chromosomal regions. These large-scale genomic differences are called **structural variants** and involve at least 50 nucleotides and as many as thousands of nucleotides that have been inserted, deleted, inverted or moved from one part of the genome to another. Tandem repeats that contain more than 50 nucleotides are considered structural variants; in fact, such large tandem repeats account for nearly half of the structural variants present in human genomes. When a structural variant reflects differences in the total number of nucleotides involved, it is called a copy-number variant (CNV). Note that CNVs are distinguished from other structural variants, such as inversions and translocations, because the latter types often do not involve a difference in the total number of nucleotides.

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



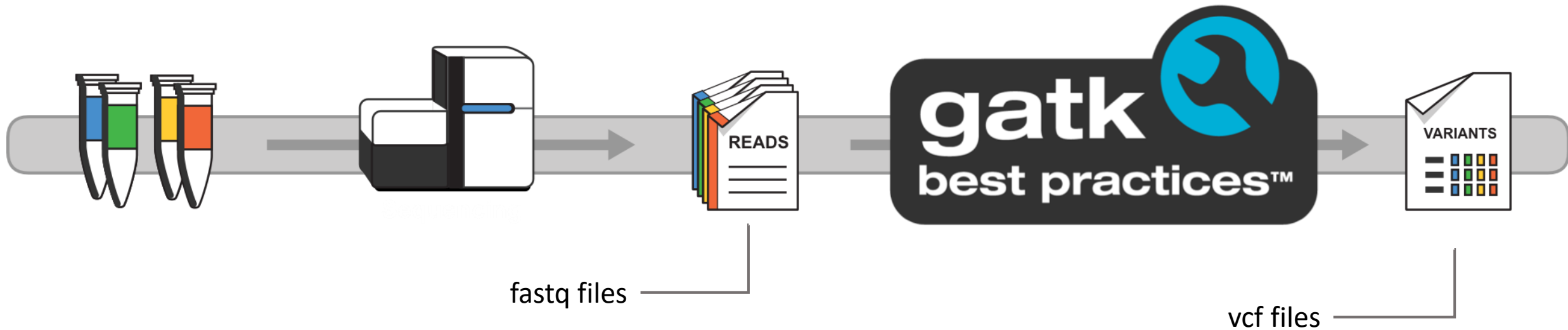
- What does a typical **human genome** sequence look like with respect to the variants that it contains? On average, compared to a reference human genome, a person's ~6 billion-nucleotide genome sequence will have:
- **~5,000,000 SNVs** that involve **~5,000,000 nucleotides**
- **~600,000 insertion/deletion variants** that involve **~2,000,000 nucleotides**
- **~25,000 structural variants** that involve **>20,000,000 nucleotides**.
- These numbers represent current estimated averages and will likely change as more human genomes are sequenced to completion. Furthermore, multiple approaches can be used to calculate how the different types of genomic variants account for the differences among genomes. In fact, calculating the total number of nucleotides involved in structural variants in each **human genome** is quite complicated and remains an active area of research.

What Is the Inventory of Genomic Variants in a Typical Human Genome Sequence?



On average, a **human genome** sequence (consisting of ~6,000,000,000 nucleotides) is ~99.6% identical — or non-variable — compared to a reference human genome sequence. The variable portion of a human genome sequence totals ~0.4%, which consists of the indicated contributions from each of the three major types of genomic variants. **These same general numbers are also found when directly comparing two peoples' genome sequences (as opposed to comparing to a reference human genome sequence).**

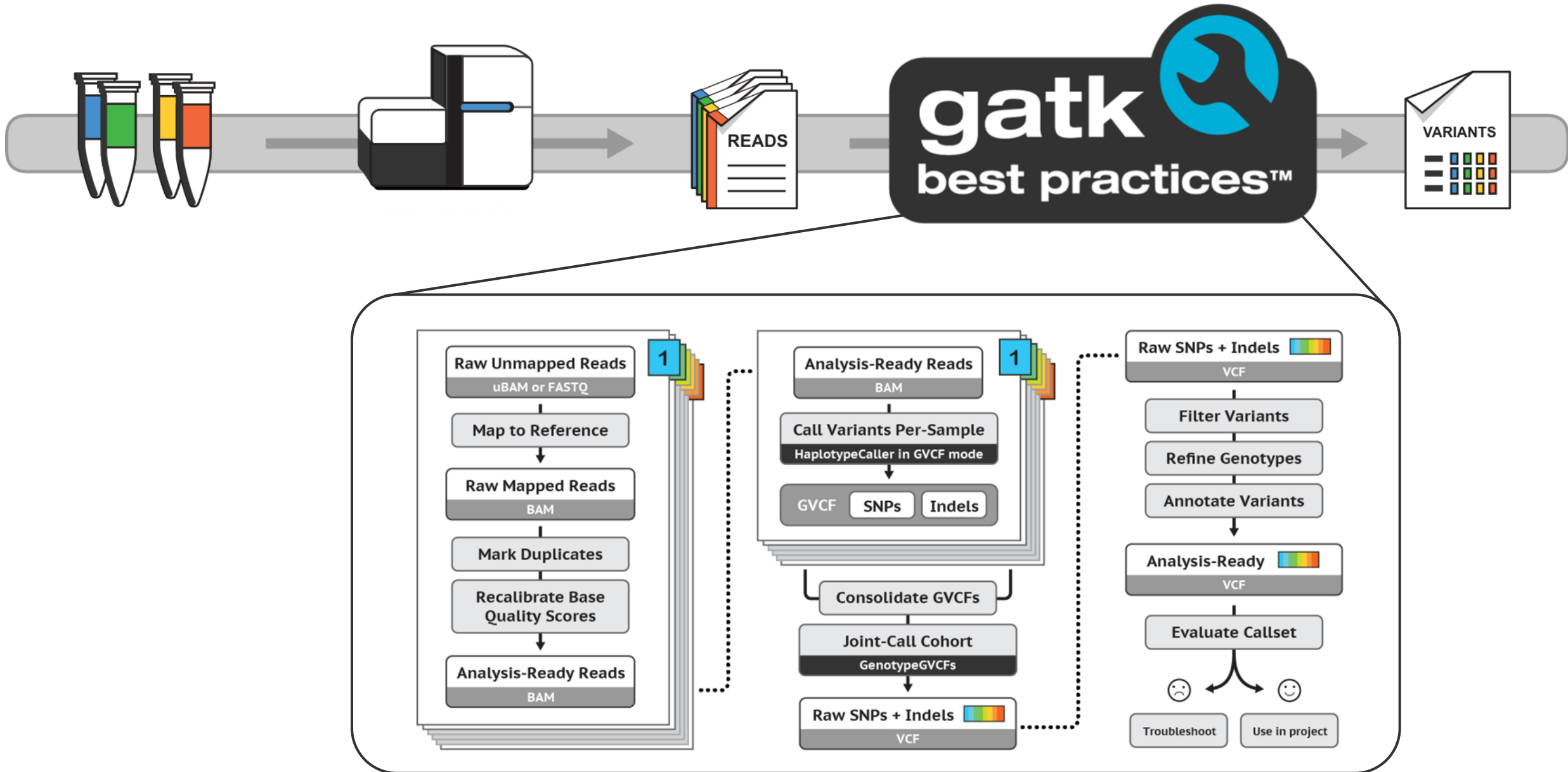
Variant discovery : identify variants in sequencing data



Variant calling pipeline : from raw sequencing reads to VCF

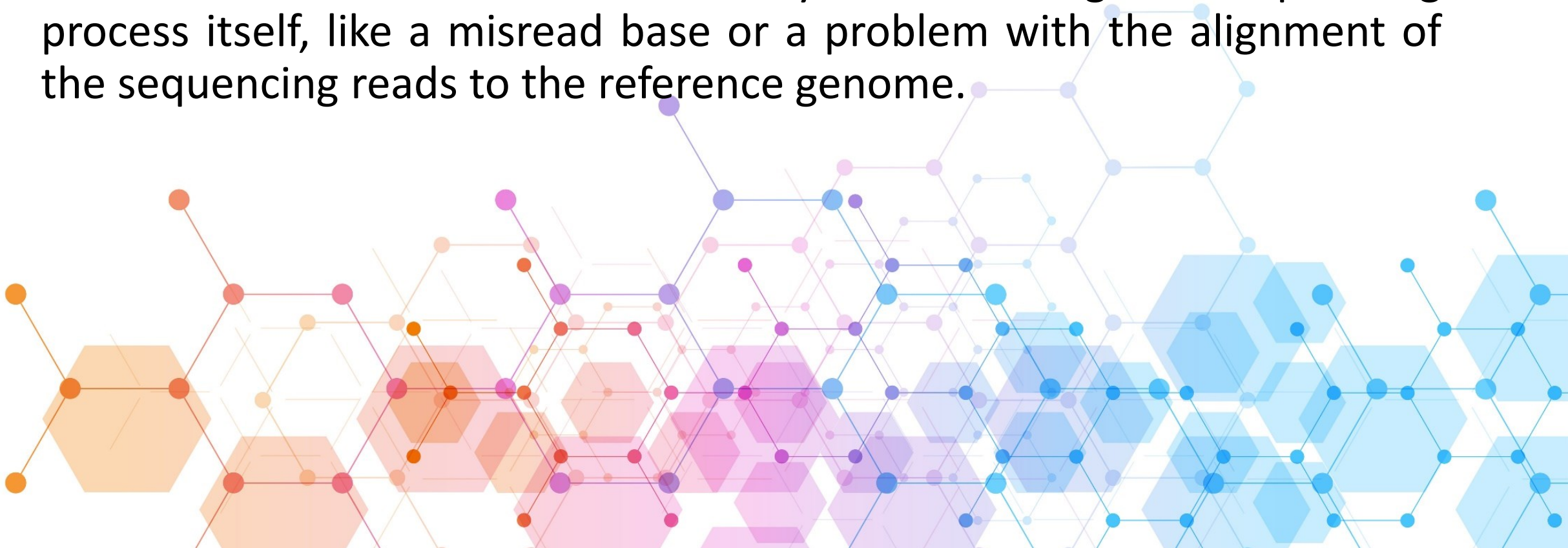
Variant discovery is the process of detecting differences between a sample genome and a reference genome

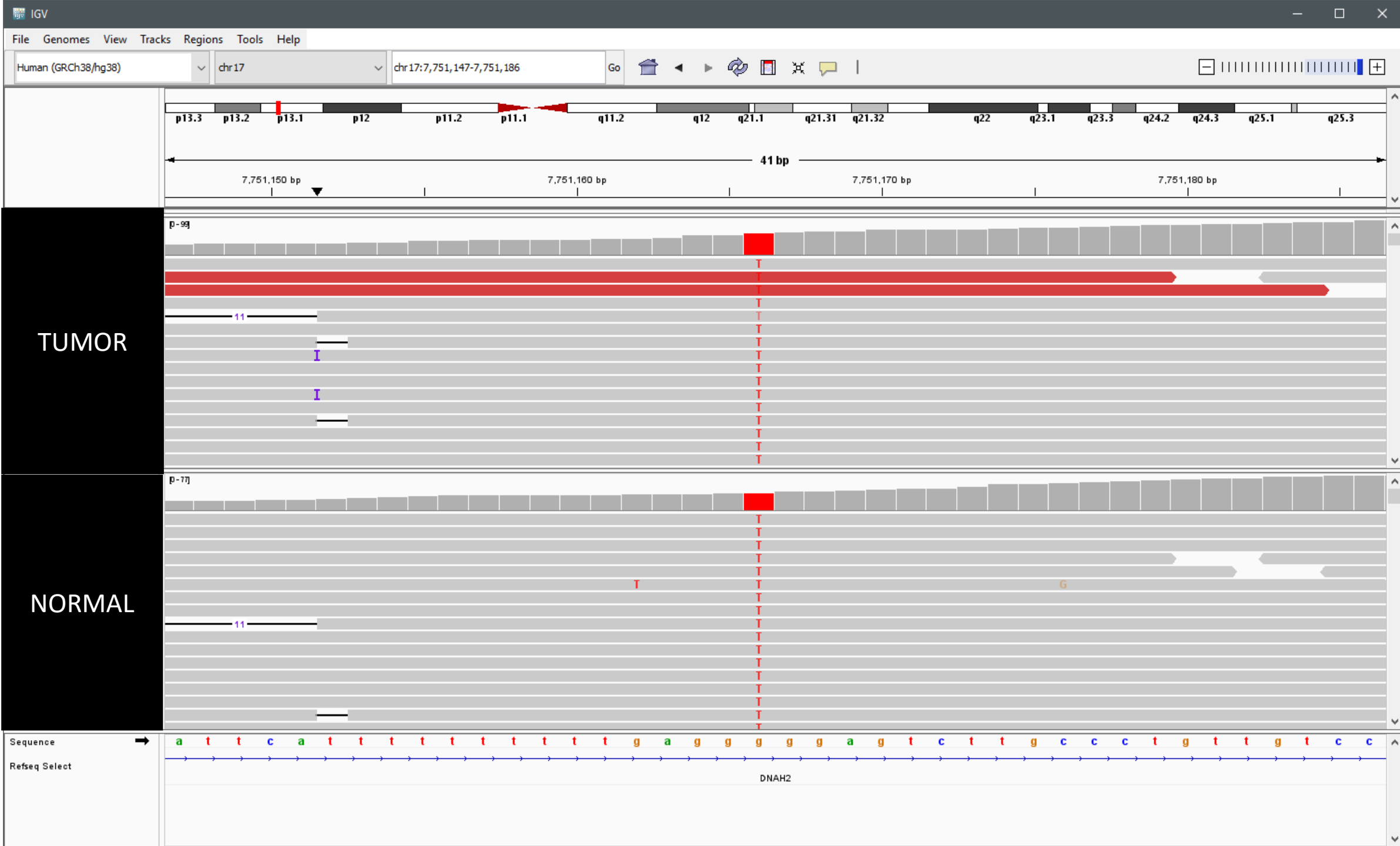
Variant discovery : identify variants in sequencing data

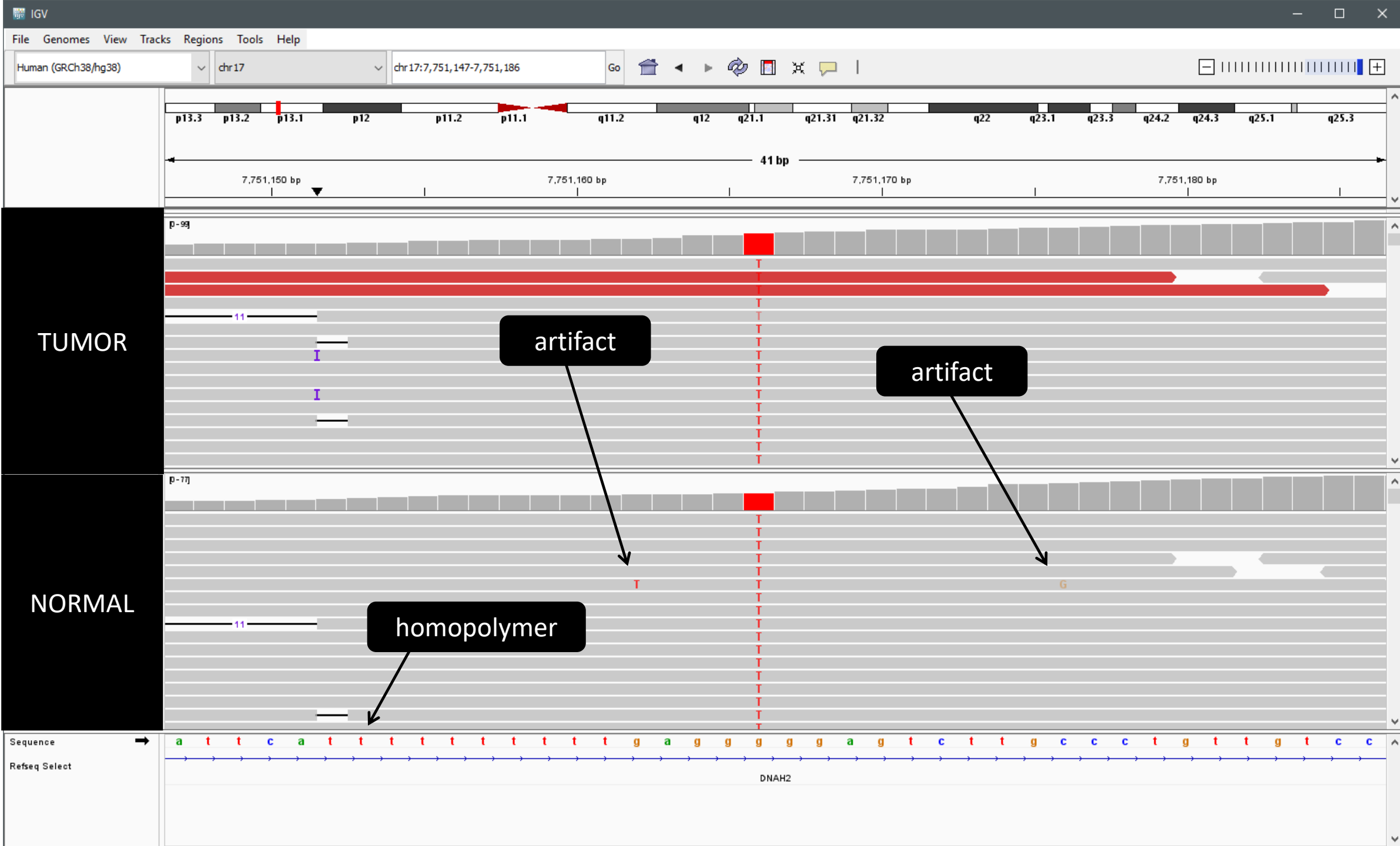


The Challenge of Distinguishing True Variants from Artifacts

A **genetic variant** is a genuine change in the DNA sequence that is actually present in the genome of the individual or a cell population. In contrast, an **artifact** is an *observed* change in the sequence data that isn't real. These are often caused by errors during the sequencing process itself, like a misread base or a problem with the alignment of the sequencing reads to the reference genome.







Common Next-Generation Sequencing Artifacts

Sequencing-Specific Artifacts

These are errors introduced during the DNA sequencing process itself. They are often platform-specific and can appear as "variant calls" at positions that aren't actually different from the reference.

- **Low-Quality Bases:** A base call with a low Phred score (e.g., $Q < 30$) has a higher chance of being a misread. These can look like a variant, but are simply an error.
- **Read-Through or Contamination:** If DNA libraries were not properly cleaned, adapter sequences can be read and incorrectly aligned, leading to false variants.
- **Strand Bias:** This occurs when a potential variant is almost exclusively observed on reads mapped to only one strand (either the forward or reverse strand). A **true variant** should be equally represented on both strands. If it appears only on one, it often indicates a technical error, such as damage to the DNA during library preparation or a systematic error during the sequencing cycle, leading to a high-confidence false positive.

Alignment-Specific Artifacts

These occur when the sequencing reads are aligned to the reference genome, even if the reads themselves are high-quality.

- **Misalignment near Indels:** Reads containing a small insertion or deletion (indel) can be difficult to align correctly. They might be shifted slightly, causing nearby bases to look like false single nucleotide variants.
- **Reads Misaligned to a Parologue or Pseudogene:** A read from a gene might incorrectly align to a similar-looking pseudogene (a non-functional copy of the gene) somewhere else in the genome. This makes the entire read look "different" and can cause a flood of false variant calls.

Low Coverage and Allele Drop-out

This is a broad category covering issues related to insufficient or uneven read depth, which can compromise variant detection.

- **Low Depth/Coverage:** If a position is covered by very few reads (e.g., < 5), the statistical power to distinguish a true variant from a random sequencing error is extremely low.
- **Allele Drop-out (ADO):** In heterozygous regions (Ref/Alt), one of the two alleles is sometimes preferentially amplified or sequenced poorly. If the Alt allele drops out, the position is falsely called Homozygous Reference (Ref/Ref). Conversely, if the Ref allele drops out, the position is falsely called Homozygous Alternate (Alt/Alt).

Understanding the Landscape of Genetic Variation

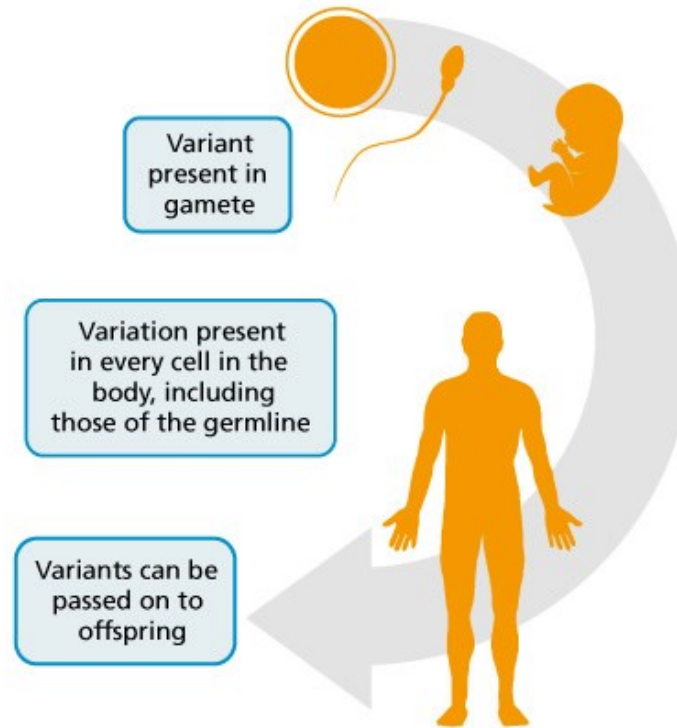
Germline vs. Somatic Variants

- **Germline Variants**

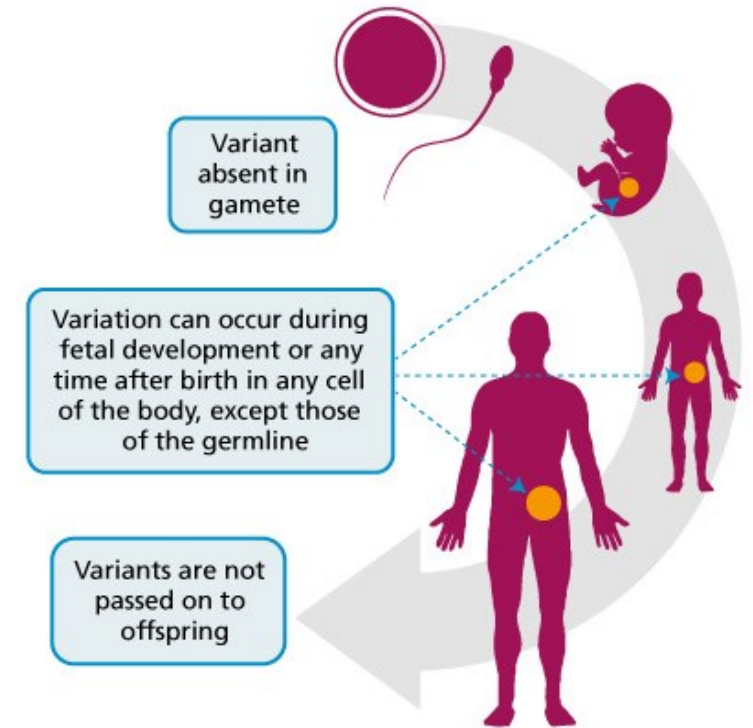
These are inherited from a parent and are present in every cell of an individual's body. They arise from genetic mutations in the egg or sperm cells. Calling germline variants is a key component of understanding inherited disease risk, population genetics, and human evolution. For example, a single nucleotide polymorphism (SNP) associated with a higher risk for a certain condition would be a germline variant.

- **Somatic Variants**

These are non-inherited genetic changes that occur after conception. They are not found in every cell of the body but are present in specific cell populations, such as a tumor. A classic example is a cancer-driving mutation like a change in the EGFR gene, which is found only in the tumor cells and not in the surrounding healthy tissue. Identifying somatic variants is the cornerstone of cancer genomics and targeted therapy.



Germline variants



Somatic variants

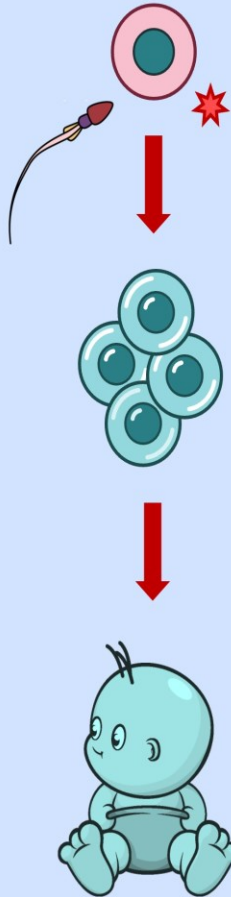
Somatic mutations vs germline mutations

SOMATIC



local effect

GERMLINE



entire organism
will be affected

Somatic mutation

It is defined as any alteration in the genetic sequence of genes of the somatic cells.

It is defined as any alteration in the genetic sequence of genes of the germinal cells.

It occurs in somatic or body cells.

It is not inheritable.

It affects only mutated cells and their progeny.

It can occur at any stage of the life cycle.

It plays no role in evolution.

It does not cause genetic disorders but may cause cancer.

It is finished along with the death of the individual.

In most cases, they show observable effects.

They can be treated or cured.

Mosaicism occurs.

Examples include cancers, tumor development, a neurodegenerative disorder, etc.

Germline mutation

It is also called “acquired mutation” as it is acquired during an individual’s life.

It is also called “hereditary mutation” as it is passed to offspring.

It occurs in germ cells.

It is inheritable.

It affects all the cells of the organism.

It occurs only during gametogenesis.

It is the basis of evolution.

It is responsible for genetic disorders and also germline cancers.

It continues till the offspring of the individual breed and may result in separate sub-species. Hence it is genetically more important.

In most cases, they are silent and don’t show detectable effects.

They can’t be treated or cured.

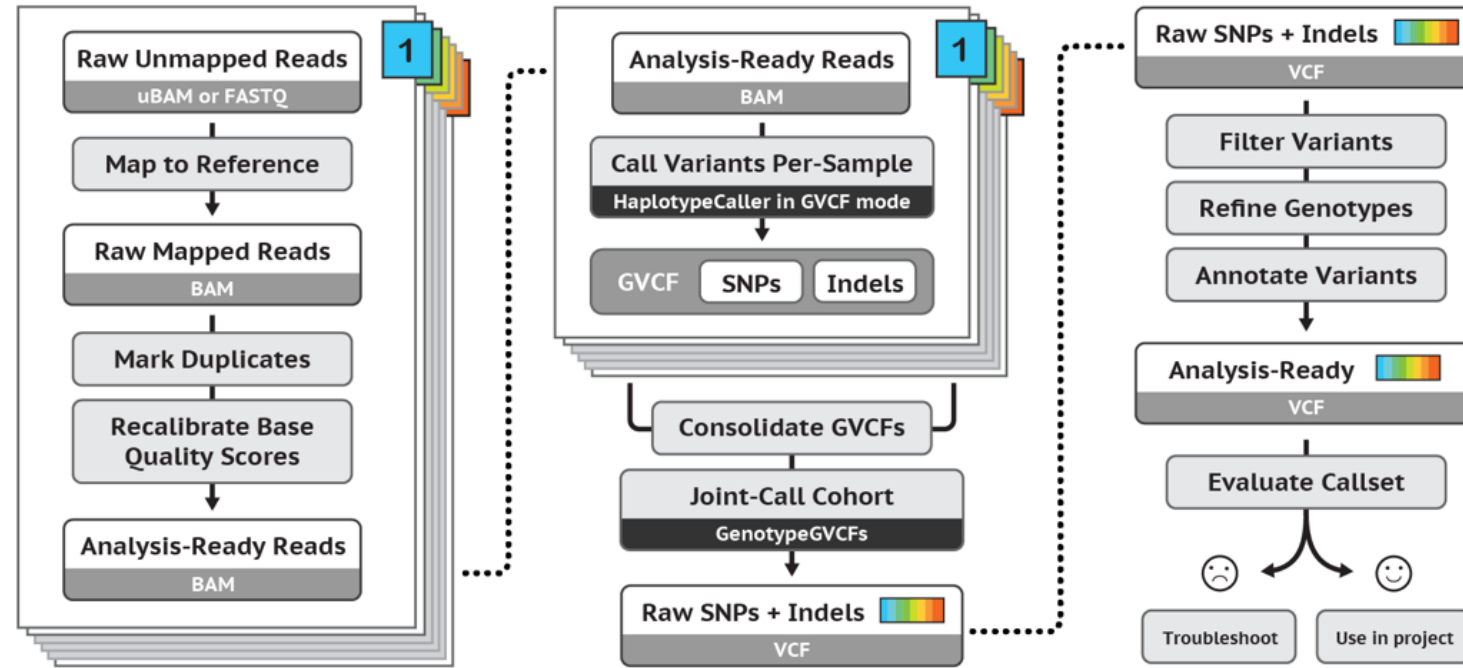
Usually, mosaicism doesn’t occur.

Examples include Hemophilia, Down’s Syndrome, 18-Trisomy, etc.



SECTION BREAK

Phases of variant calling



1. Sequence
data
processing

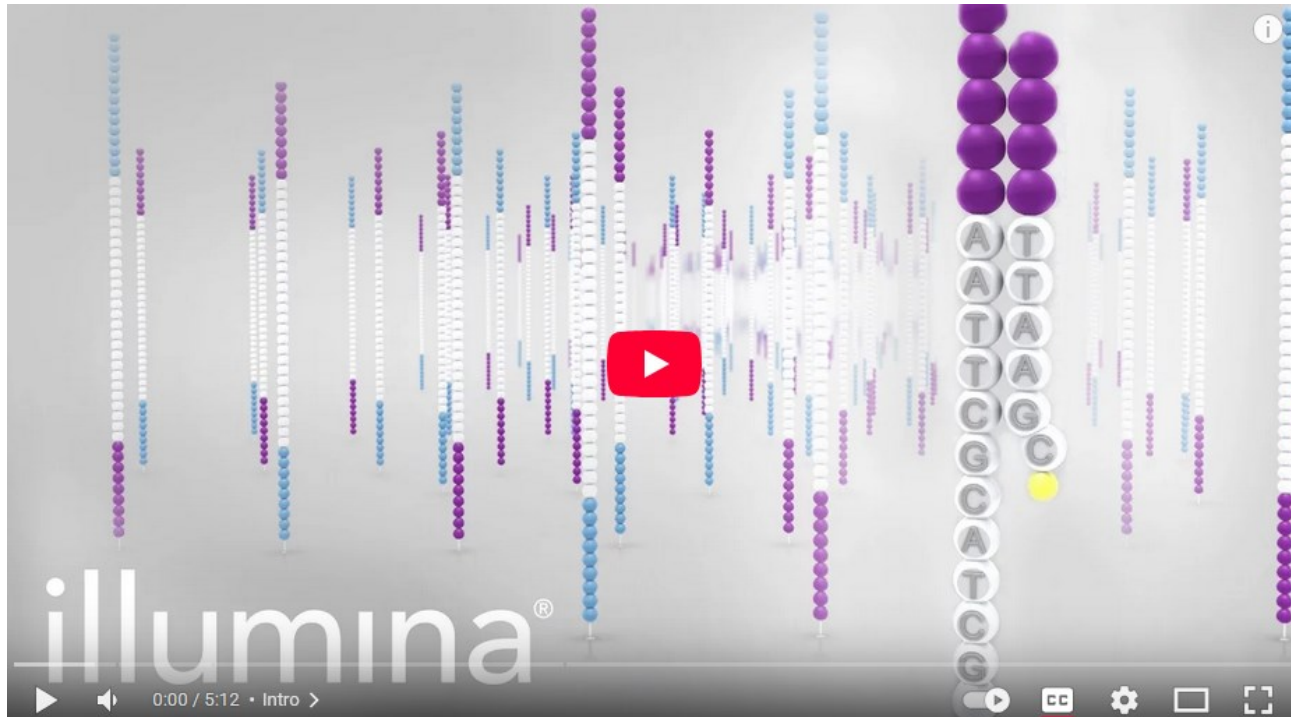
2. Variant
calling

3. Variant
filtering

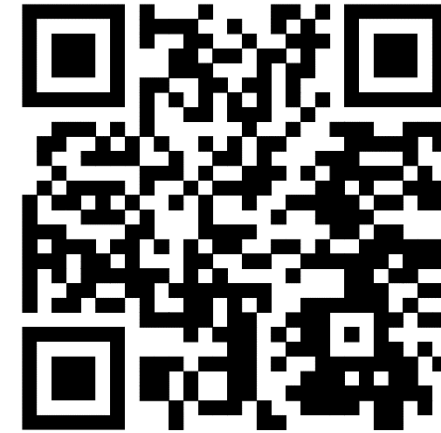
4. Variant
annotation

5.
Evaluation
& QC

Sequencing by synthesis



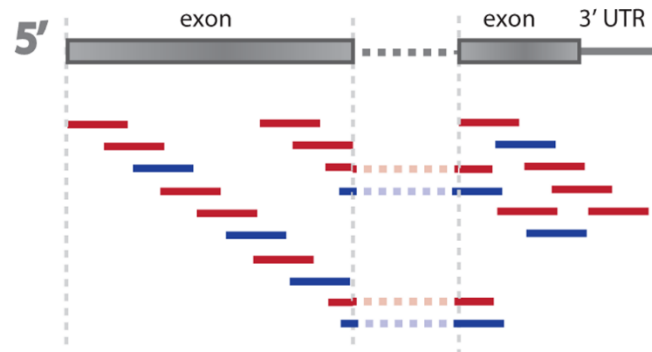
<https://www.youtube.com/watch?v=fCd6B5HRaZ8>



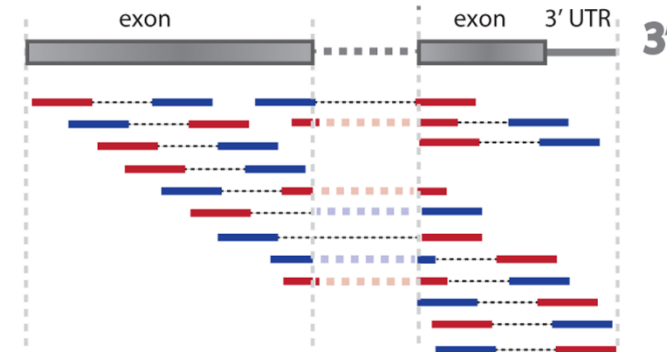
Overview of Illumina
Sequencing by Syn-
thesis

Single-end/Paired-end sequencing

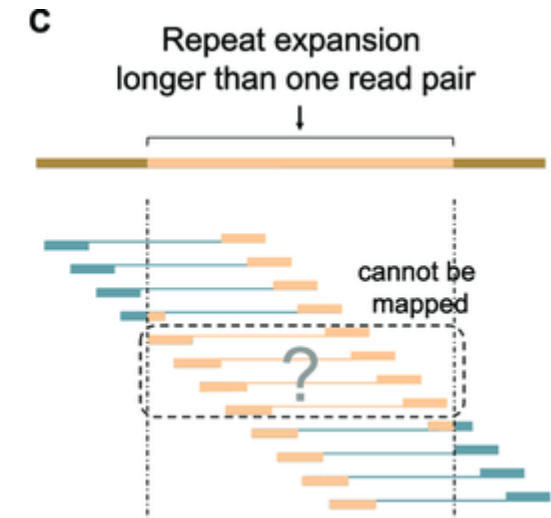
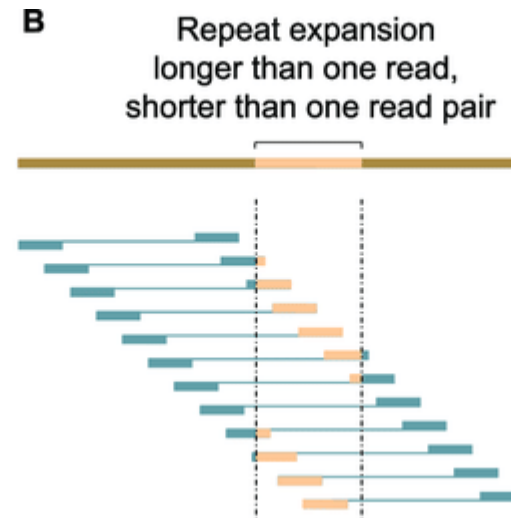
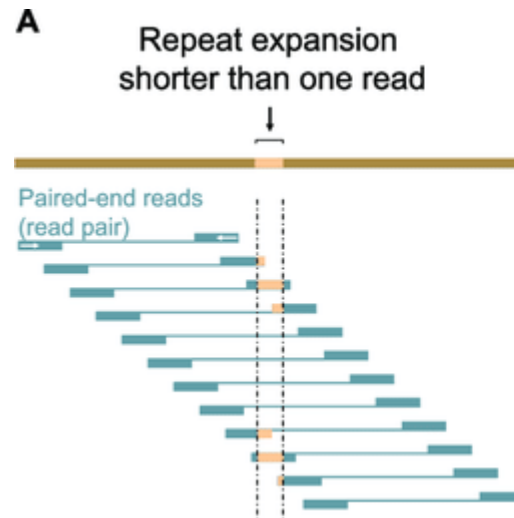
Single-end sequencing

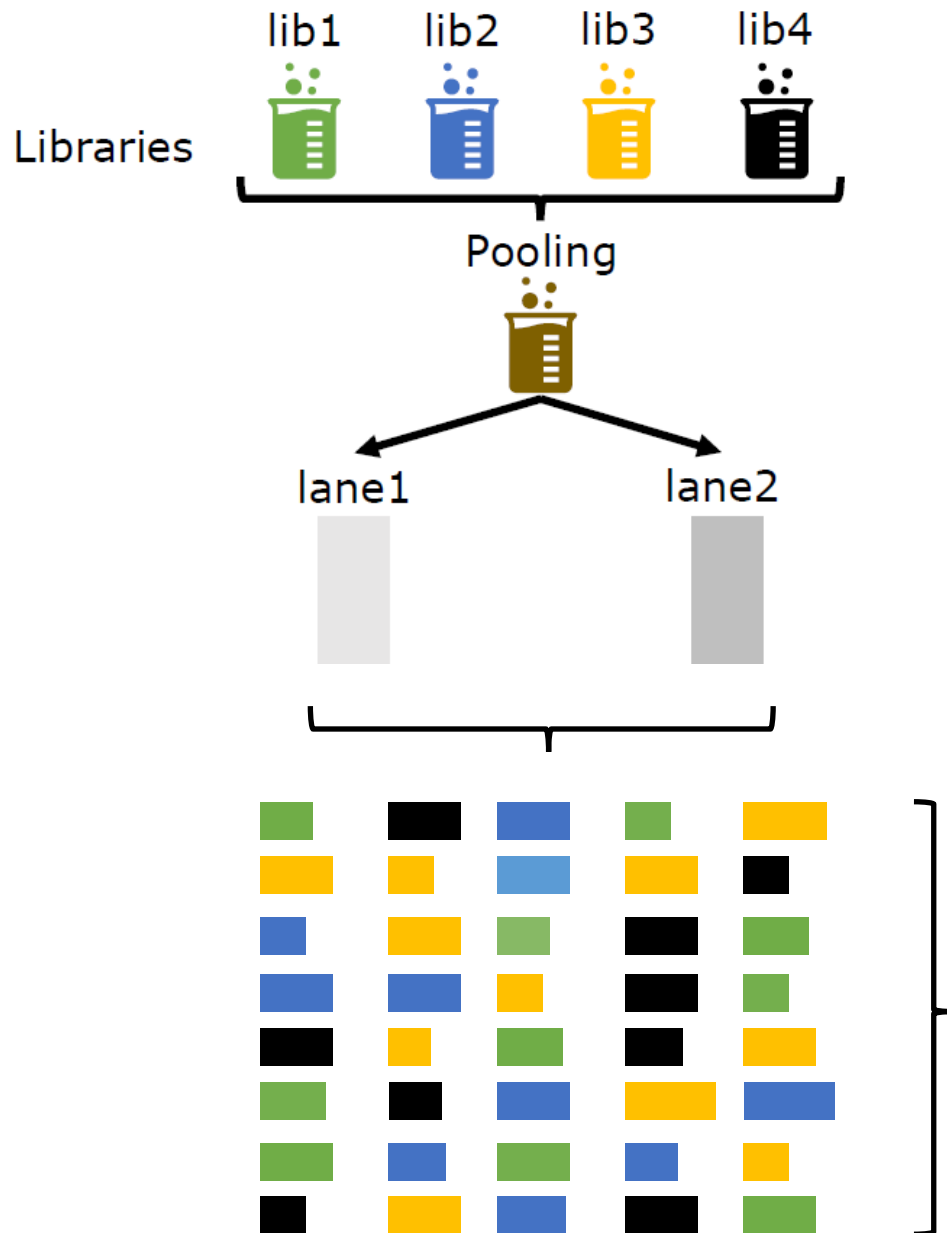


Paired-end sequencing



Paired-end gives us more information, especially useful in detecting indels or structural changes.





Multiplexing/Demultiplexing

Pooling (Multiplexing)

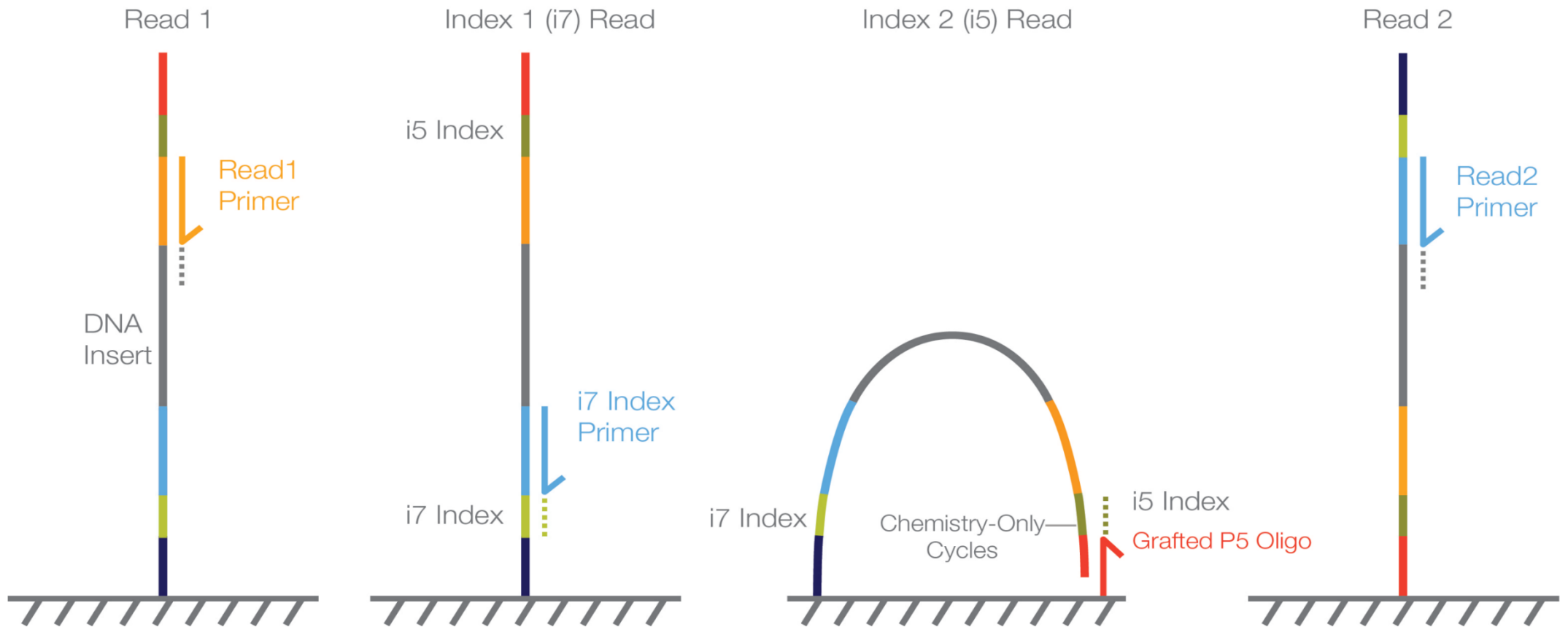
Combining multiple DNA libraries (from different samples) into a **single sequencing run**.

Each sample is prepared with a **unique index (barcode)** embedded in the adapters (usually in the P7/P5 adapters).

Demultiplexing (splitting the reads)

Sorting reads from the pooled FASTQ into separate files **based on the index sequences**.





Adding unique barcodes to the index reads allows different samples to be sequenced concurrently. Following this, the unique index reads are used to separate out the original samples.

Raw Sequencing Reads (FASTQ)

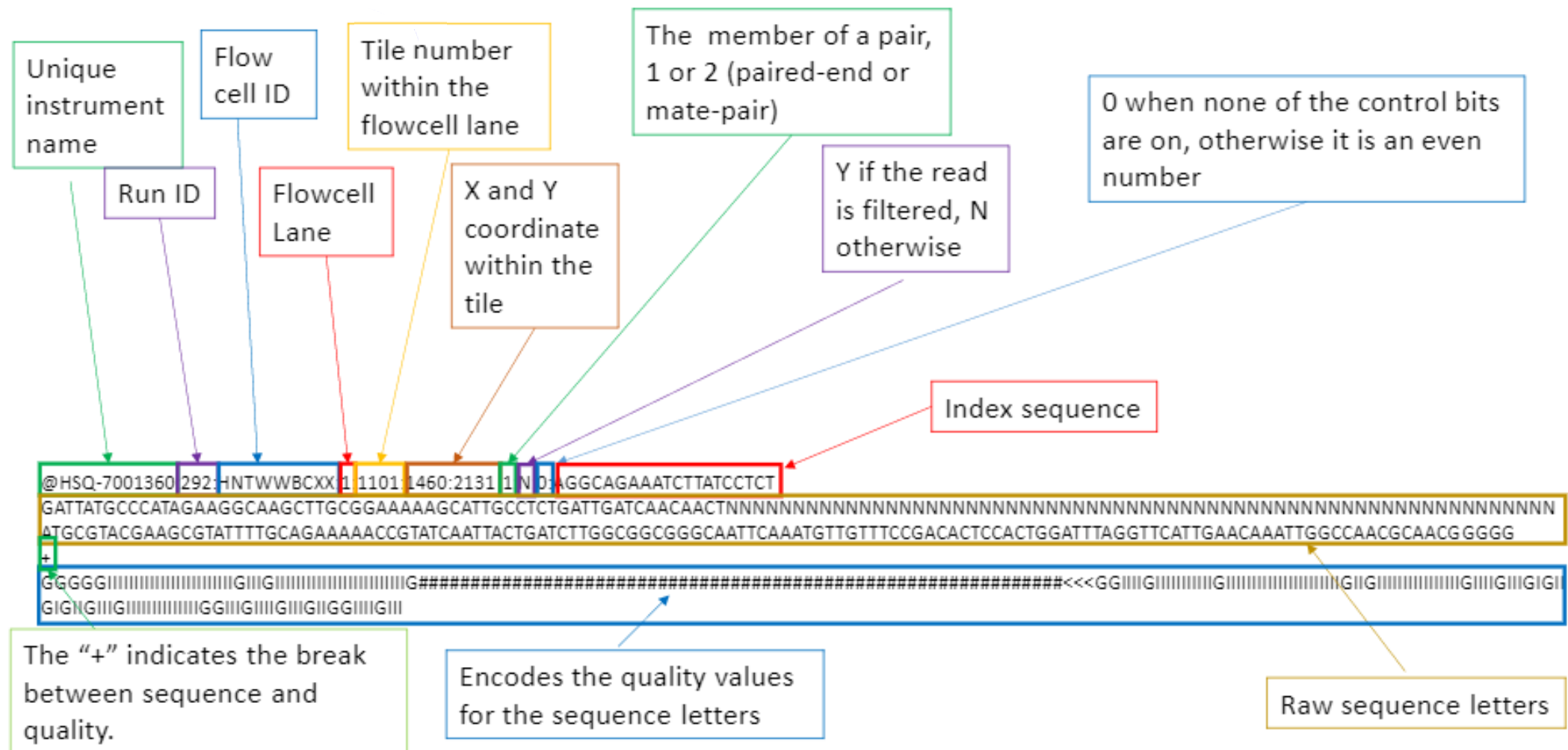
```

identifier
sequence
+
quality
identifier
sequence
+
quality

```

```
@M00877:474:00000000-K7HDK:1:1101:10371:2108 1:N:0:TCTCATGA+CTTCTAAT  
TTTATGGCCTACGGGAGGCAGCAGTGAGGAATTATTGGACAATGGAGGCCAACTCTGATCCTGCCATTCCGCGTGTTTTGTCTACGTCCCCTATGTTCTTTTTTCTTCTTTTTTCTCTG,  
+  
-ACCCGGGGGGGGGGGGGGGGGGGGGGCFGF9FFAFDEF8EFE<A,,8,,:CFF,,,6,,,,,<,6+8+, ,,,,: ,,,66:, : ,,,<,9,,+495,CCE,,+4,,.,.  
@M00877:474:00000000-K7HDK:1:1101:10833:2162 1:N:0:TCTCATGA+CTTCTAAT  
TATTAAGCCTACGGGCGGCAGCAGTGAGGAATTATGCACAATGGGCGCAAGCCTGATGCTGCCATGCCGCGTGATGAAGAAGGCCTTCTGTTTTTTAAGTACTTTCTTCTTTGAI  
+  
@ACCCGGGGGGGGGGDGGGGGGGGGGGGGGGGGGGGGGGGGGGGEGEC@@@76DF8,C,6,,C6C?,6C+7+8+B,C,,,,,,,,,9:?E,,C?,D+, ,,: :CEF,,,,,,,,..  
@M00877:474:00000000-K7HDK:1:1101:16190:2217 1:N:0:CCTCATGA+CGTCTAAT  
CGCTAGGCCTACGGGAGGCAGCAGTGAGGAATTATGCACAATGGGCGAAAAGCCTGATGCAGCAAACGCCGCTTAGCGATGAAGGCCTTCTGTTTCGTTAAGCTCTTTCCTCAAGGA,  
+  
@BCCCCGGGGGGGGGGGGGGGGGGFGGGGGGGGGGGGGGGGGGGGD77+@C<FGD8CC<,9C,,::++@@+@+8+, ,4+: ,,,444C=F,, ,C9+B,++,5AED,B55AA,,,,,,,,.  
@M00877:474:00000000-K7HDK:1:1101:20697:2220 1:N:0:GCTCATGA+CGTCTAAT  
TGACCTACCTACGGGAGGCTGCAGTGAGGAATTATGGTCAATGGTCGCGAGACTGATCCTGCCCTGTAGCGTTCATGTCGACTGCCCTCCGTTTTTTTTTCTGCTTTTTTTTTTGT'  
+
```

Raw Sequencing Reads (FASTQ)

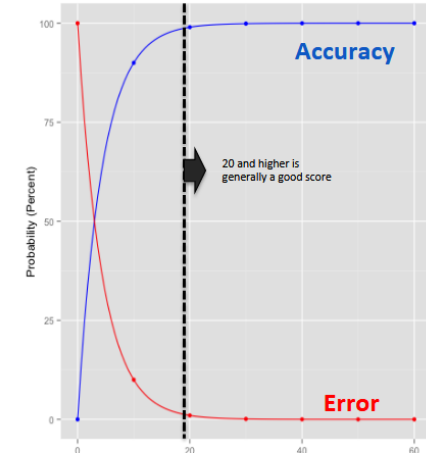


Quality (Phred) scores

Phred value = $-10 \cdot \log_{10}(\text{err})$

Example:

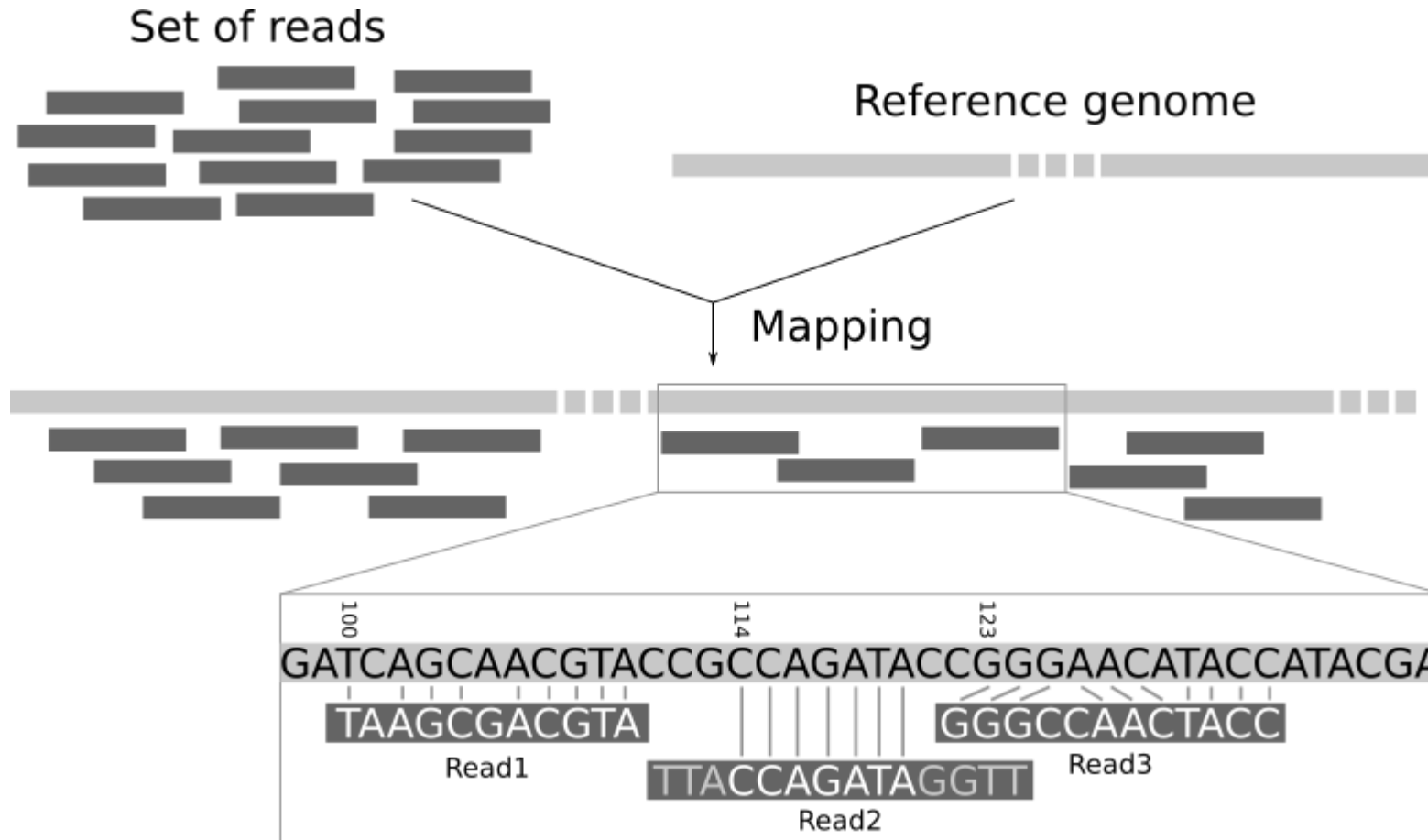
- 90.0% confidence (10% error rate) = Q10
- 99.0% confidence (1% error rate) = Q20
- 99.9% confidence (0.1% error rate) = Q30



Illumina v1.8 and later (ASCII_BASE=33)

Q	ASCII	P	Q	ASCII	P	Q	ASCII	P	Q	ASCII	P
1	"	0.79433	12	-	0.06310	23	8	0.00501	34	C	0.00040
2	#	0.63096	13	.	0.05012	24	9	0.00398	35	D	0.00032
3	\$	0.50119	14	/	0.03981	25	:	0.00316	36	E	0.00025
4	%	0.39811	15	0	0.03162	26	;	0.00251	37	F	0.00020
5	&	0.31623	16	1	0.02512	27	<	0.00200	38	G	0.00016
6	'	0.25119	17	2	0.01995	28	=	0.00158	39	H	0.00013
7	(	0.19953	18	3	0.01585	29	>	0.00126	40	I	0.00010
8	)	0.15849	19	4	0.01259	30	?	0.00100	41	J	0.00008
9	*	0.12589	20	5	0.01000	31	@	0.00079			
10	+	0.10000	21	6	0.00794	32	A	0.00063			
11	,	0.07943	22	7	0.00631	33	B	0.00050			

2. Alignment to Reference Genome (BAM/CRAM)



- Align short reads to a known reference (e.g., GRCh38).
- Tools: BWA-MEM, Bowtie2, minimap2.
- Output: SAM/BAM/CRAM file (binary alignment).
- Why? To know where each read comes from in the genome.

Challenges:

- Reads mapping equally well to multiple locations.
- Indels near repetitive regions.

131	141	151	161	171	181	191	
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGGC							← reference
.....							
CATC		CCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGC				CTGGC	
CATCTT		TCCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTC				(	
CATCTTCAACACCCGCC		CCCGCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCC					
CATCTTCAACACCCGCCTCT		GCACCTTCGGATATACTATCAAGCGAACCACAGTCAAAACGCCCTCCTGG					
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATAT				TCAAGCGAACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTA				CGAACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAA				ACCACAGTCAAAACGCCCTCCTGGC			
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA				ACAGTCAAAACGCCCTCCTGGC			
CATCTTCAA					GTCAAAACGCCCTCCTGGC		
CATCTTCAACACC						CCCTCCTGGC	
CATCTTCAACACCCGCCTCTC							
CATCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAA							
	TCTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACC						
	CTTCAACACCCGCCTCTCCCGCACCTTCGGATATACTATCAAGCGAACCA						← read

SAM/BAM file format

Header

Reads

Example BAM file

```
@HD VN:1.5 SO:coordinate
@SQ SN:chr1 LN:248956422
@SQ SN:chr2 LN:242193529
...
@RG ID:Case_003_T LB:Case_003_T PL:ILLUMINA SM:Case_003_T
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem -t 10 -M -R ...
Read1 99 chr1 10147 40 10M = 10395 310 ACCCTAACCC CHJ<GGIF;G RG:Z:Case_003_T
Read2 83 chr1 10178 22 7M6H3M = 10201 -65 CCCTATCCCG F=(2B9DB9) RG:Z:Case_003_T
```

Annotations:

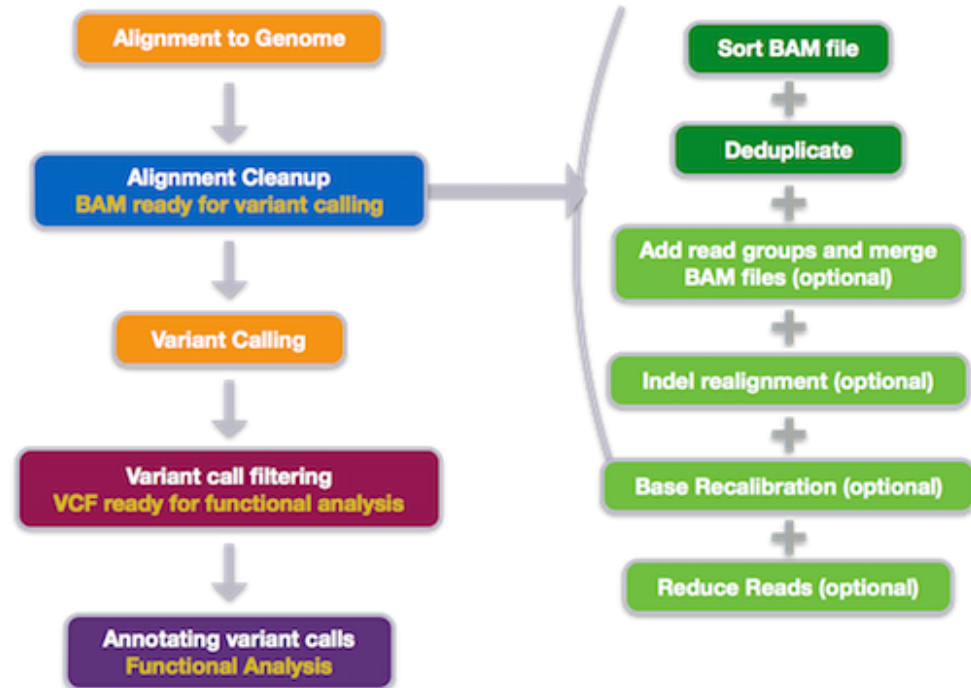
- Header line
- Reference sequence name
- Sorting order of alignments: coordinate
- Reference sequence length
- @RG: Read group
ID: Read group identifier
LB: Library
PL: Platform
SM: Sample
- @PG Program
ID: Program record identifier
PN: Program name
VN: Program version
CL: Command line.
- Read name
- Position
- Mapping quality
- Sequence
- Base quality

Base quality scores are per-base estimates of error emitted by the sequencing machines.
Short variant calling algorithms rely heavily on base quality scores

Preprocessing

Before calling variants, BAM files are *cleaned*:

- **Sorting:** order reads by coordinate.
- **Marking duplicates:** PCR duplicates can bias allele counts.
- **Base Quality Score Recalibration (BQSR):** adjust quality scores based on systematic errors.



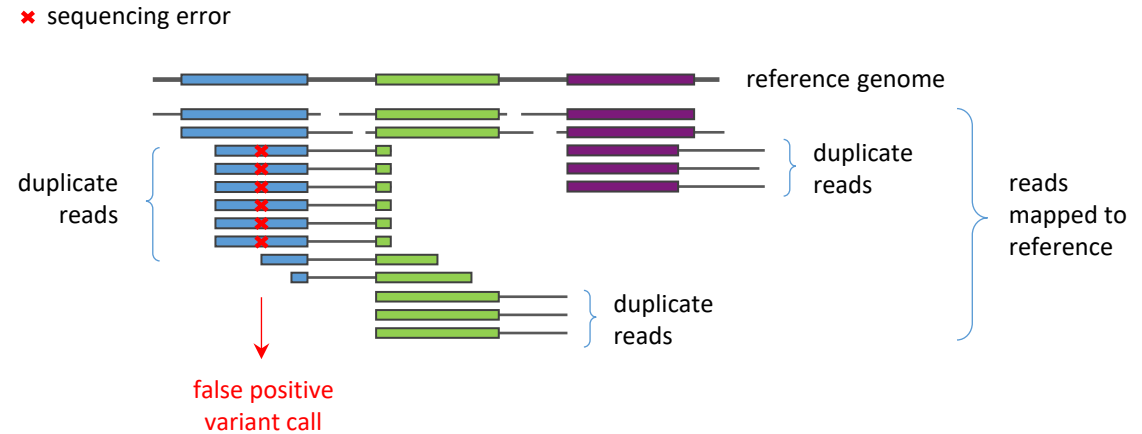
Marking/removing duplicates

Duplicate reads = multiple reads from the same original DNA fragment

- **PCR duplicates:** created during library amplification
- **Optical duplicates:** same cluster mistakenly counted more than once

Why remove them?

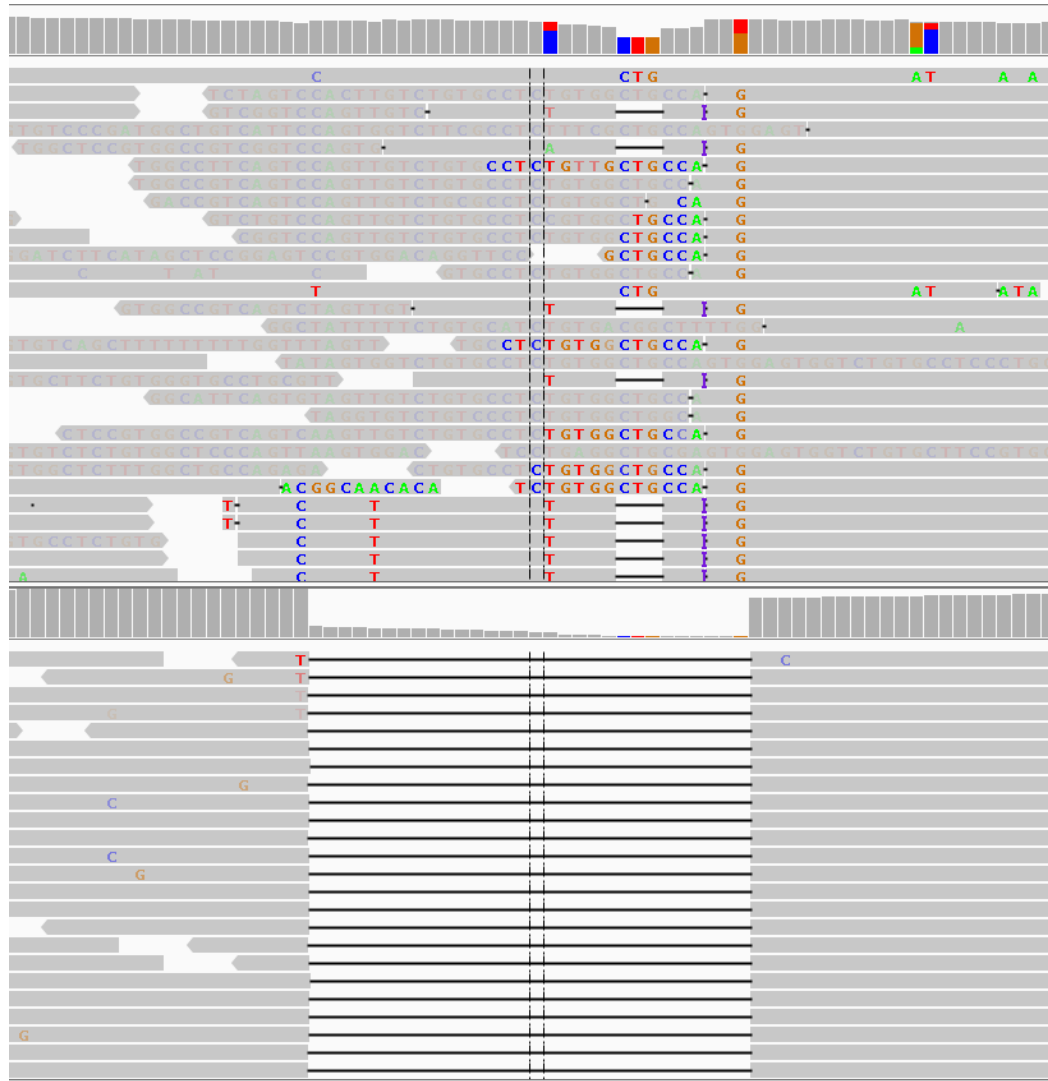
- Variant calling assumes each read is an independent observation
- Duplicates share the same origin → not independent evidence



After flagging duplicates, it is more likely that the right call will be made



Indel realignment



Before indel realignment

misaligned reads around
insertions/deletions → false variant calls

After indel realignment

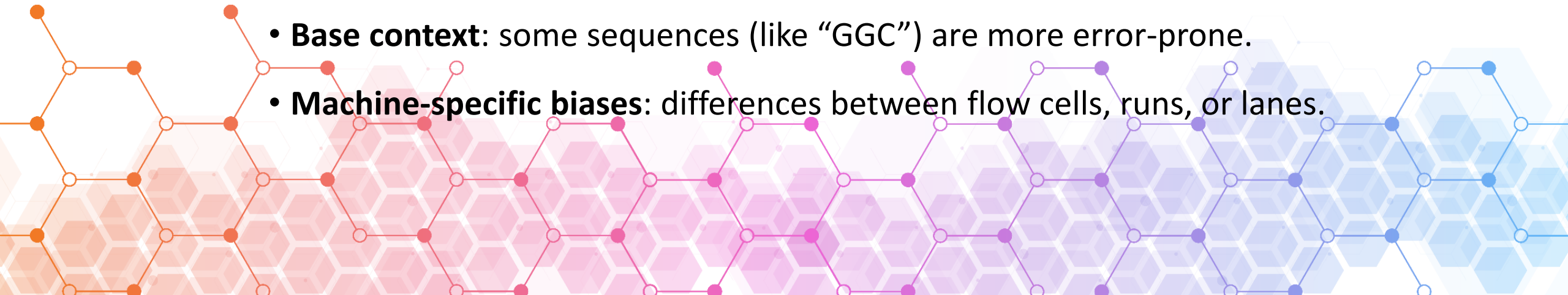
reads are properly aligned → more accurate
variant calling

Base quality score correction

- Each base in a sequencing read has a **quality score (Phred score)**
- Reflects the **probability that the base was called incorrectly**
- Essential for estimating genotype likelihood

Sequencers can make **systematic errors** depending on:

- **Cycle number:** the quality may drop toward the end of the read.
- **Base context:** some sequences (like “GGC”) are more error-prone.
- **Machine-specific biases:** differences between flow cells, runs, or lanes.

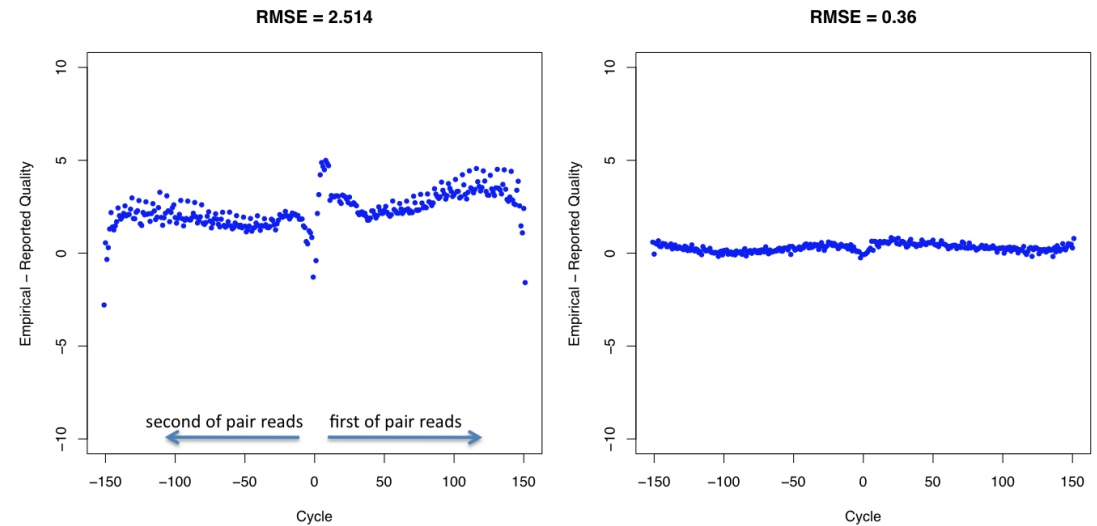


Base Quality Score Recalibration (BQSR)

Base Quality Score Recalibration (BQSR)

- Adjusts sequencer-reported base quality scores to match real error rates
- Corrects for systematic biases: read cycle, sequence context, machine-specific effects
- Makes variant calling more accurate by trusting high-quality bases and down-weighting error-prone ones

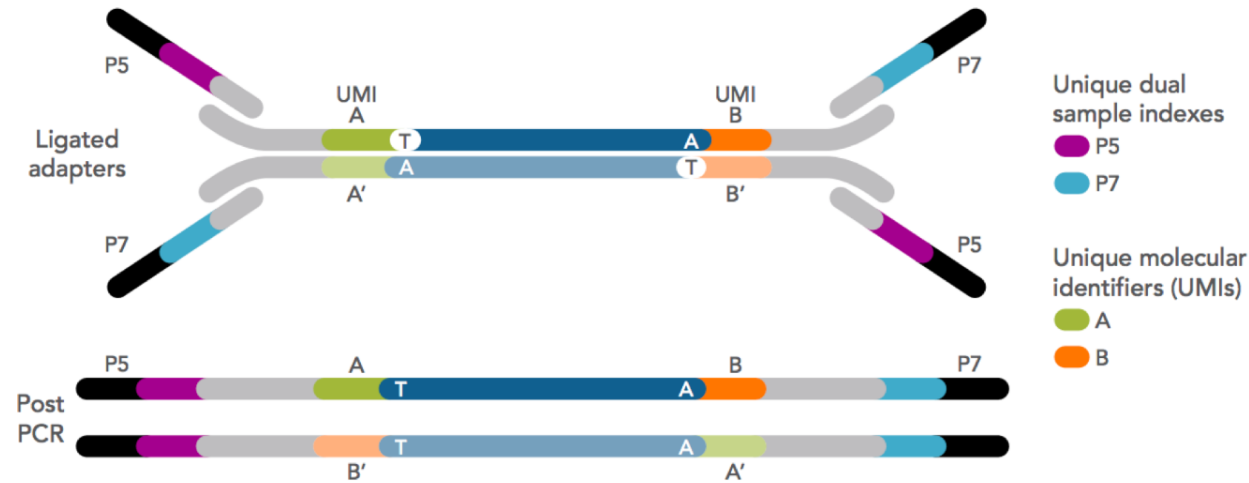
Residual Error by Machine Cycle



Original data

After
recalibration

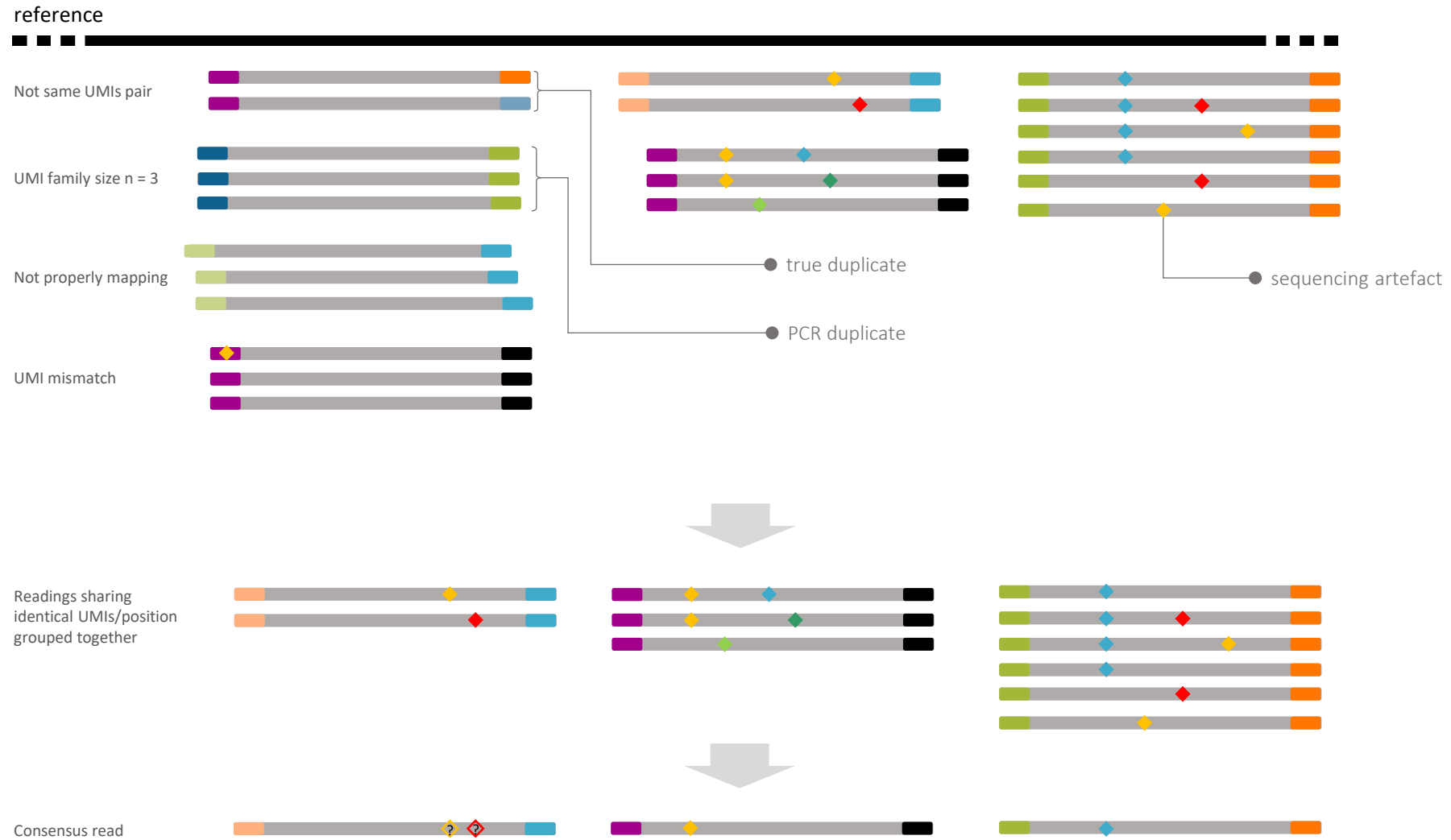
UMIs (Unique Molecular Identifiers)



UMIs – Unique Molecular Identifiers

- Short, random sequences added to each DNA/RNA molecule before amplification
- Allow true duplicate identification and consensus calling
- Enable accurate detection of low-frequency variants (e.g., cancer, liquid biopsy)

Consensus read / error correction

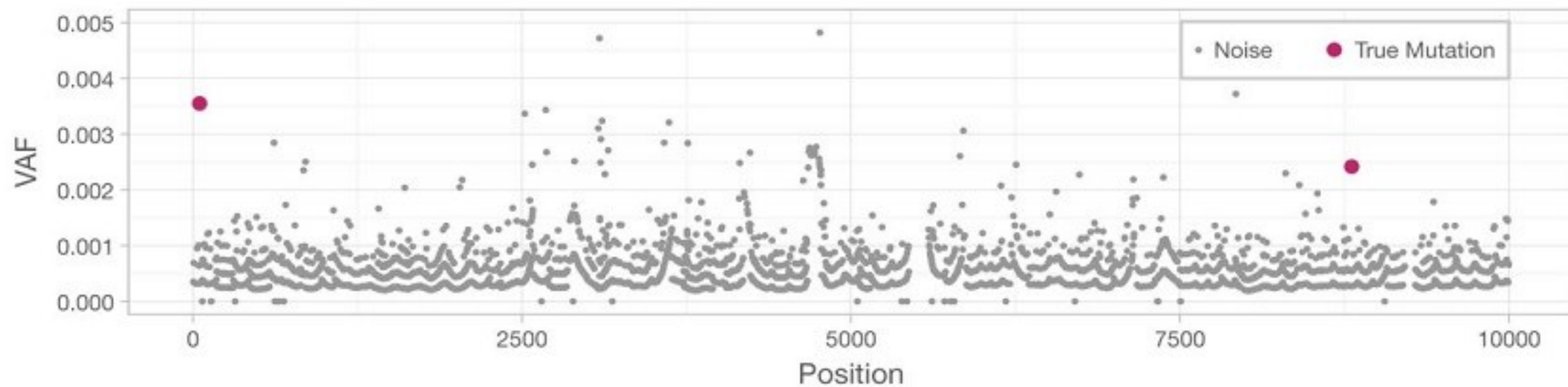


Umi pipeline

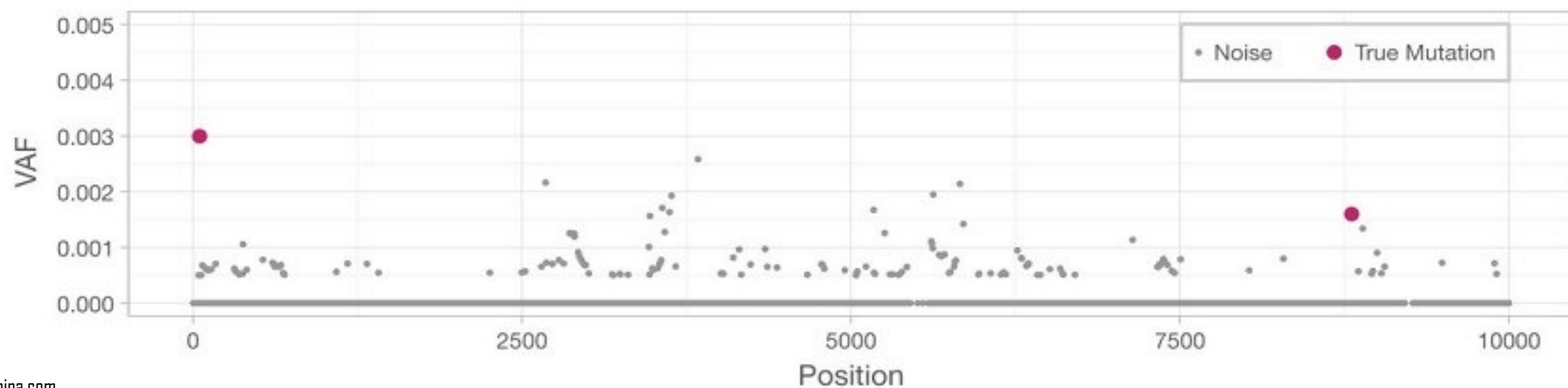
UMI pipeline



Without UMI



With UMI



4. Variant Calling

Questions

1. Is there a variant at location X?
Deviation from REF in the alignments
2. What are the alleles?
The variation in sequence in these deviations
3. What is the genotype (HomRef, Heterozygote or HomAlt)?
Estimating the allele counts in the sample
4. What is the allele frequency?
Relative proportion of the alt allele



The Statistical Nature of Variant Calling

Variant calling is fundamentally a statistical process, not a simple pattern match. It's not enough to find a mismatch between the sequenced DNA and the reference genome; the challenge lies in evaluating the evidence for that mismatch and deciding whether it represents a genuine variant or is merely a sequencing artifact. This probabilistic approach is critical because all sequencing technologies have an inherent error rate.

The process relies on three primary sources of evidence to distinguish signal from noise:

- **Sequencing Depth (Coverage)**

This refers to the number of times a specific position in the genome has been sequenced. It's a key measure of data quality. **A higher coverage provides stronger statistical evidence for a variant.** For example, if you see a mismatch at a position that has been sequenced only twice, the confidence is low. If that same mismatch appears in 100 out of 100 reads, the confidence is exceptionally high.

- **Base Quality Scores**

Each base call from a sequencer comes with a **Phred score**, a **logarithmically scaled measure of the confidence** in that specific base call. A score of Q30 means there's a 1 in 1,000 chance of the base being wrong. A variant call that is supported by reads with high base quality scores is much more trustworthy than one supported by reads with low-quality scores.

- **Allele Frequency**

This is the **proportion of sequencing reads that show the variant allele at a given position**. In a standard germline sample, this value is expected to be close to 0% (reference allele), 50% (heterozygous variant), or 100% (homozygous variant). In a tumor sample, however, the allele frequency can be much lower due to the **presence of normal cells and sub-clonal populations of cancer cells**.

Germline Variant Calling

In **germline variant calling**, the primary goal is to identify a change that is **present in all**, or nearly all, of the sequenced cells. The expectation is that the variant will be present in roughly **50%** of the reads for a **heterozygous variant** (one copy of the allele) or **100%** of the reads for a **homozygous variant** (two copies of the allele).

- **Input Data:** Typically uses DNA from a single healthy tissue sample (e.g., blood).
- **Core Assumption:** The variants are constitutional, meaning they are present in all cells.
- **Example Tool:** GATK's HaplotypeCaller is a widely used algorithm specifically designed for this purpose.

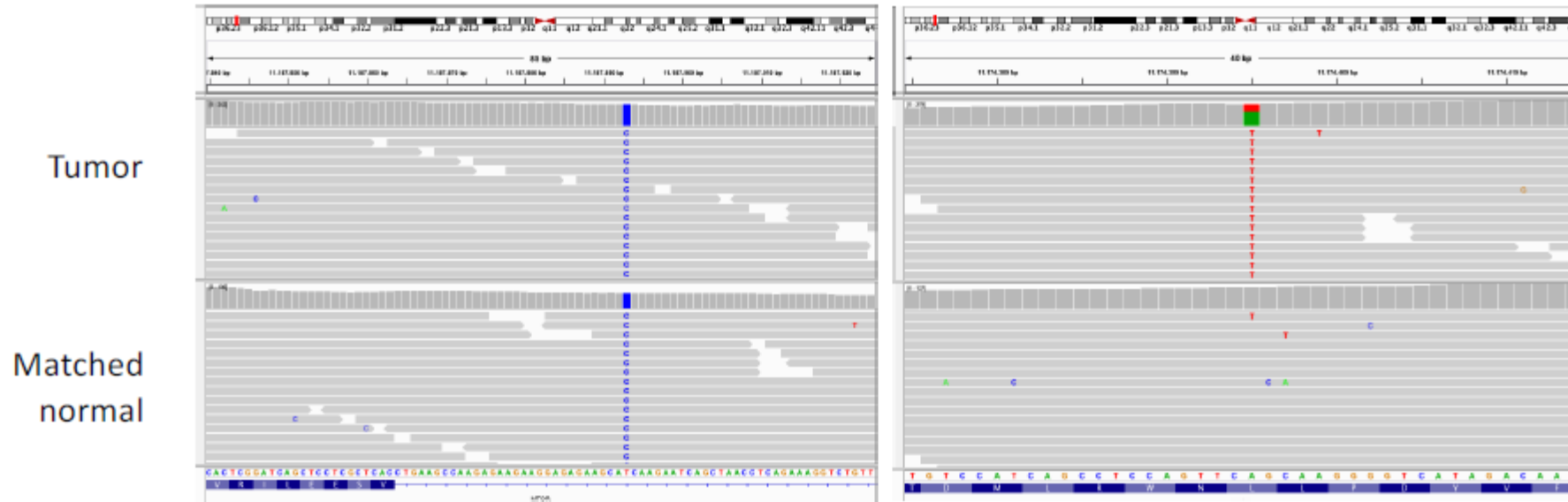


Somatic Variant Calling

Somatic variant calling is significantly **more complex** because it's looking for **changes** that are **present in only a subset of the sequenced cells**. For example, in a tumor sample, the cancer cells with the mutation are mixed with normal, non-cancerous cells. This means the variant allele frequency can be very low, well below the 50% or 100% threshold of germline variants.

- **Input Data:** Better using matched "**tumor-normal**" pair. The tumor sample is the primary source of the cancer variants, while a normal sample (e.g., blood or adjacent tissue) from the same individual serves as a crucial control to filter out germline variants.
- **Core Assumption:** Variants are sub-clonal and may be present at very low frequencies, so the signal must be distinguished from random sequencing noise.
- **Key Challenge:** Differentiating a low-frequency somatic variant from a low-quality sequencing artifact is a major statistical hurdle.
- **Example Tool:** Mutect2 is a common algorithm designed to handle this complexity by comparing the tumor and normal samples to identify changes unique to the tumor.

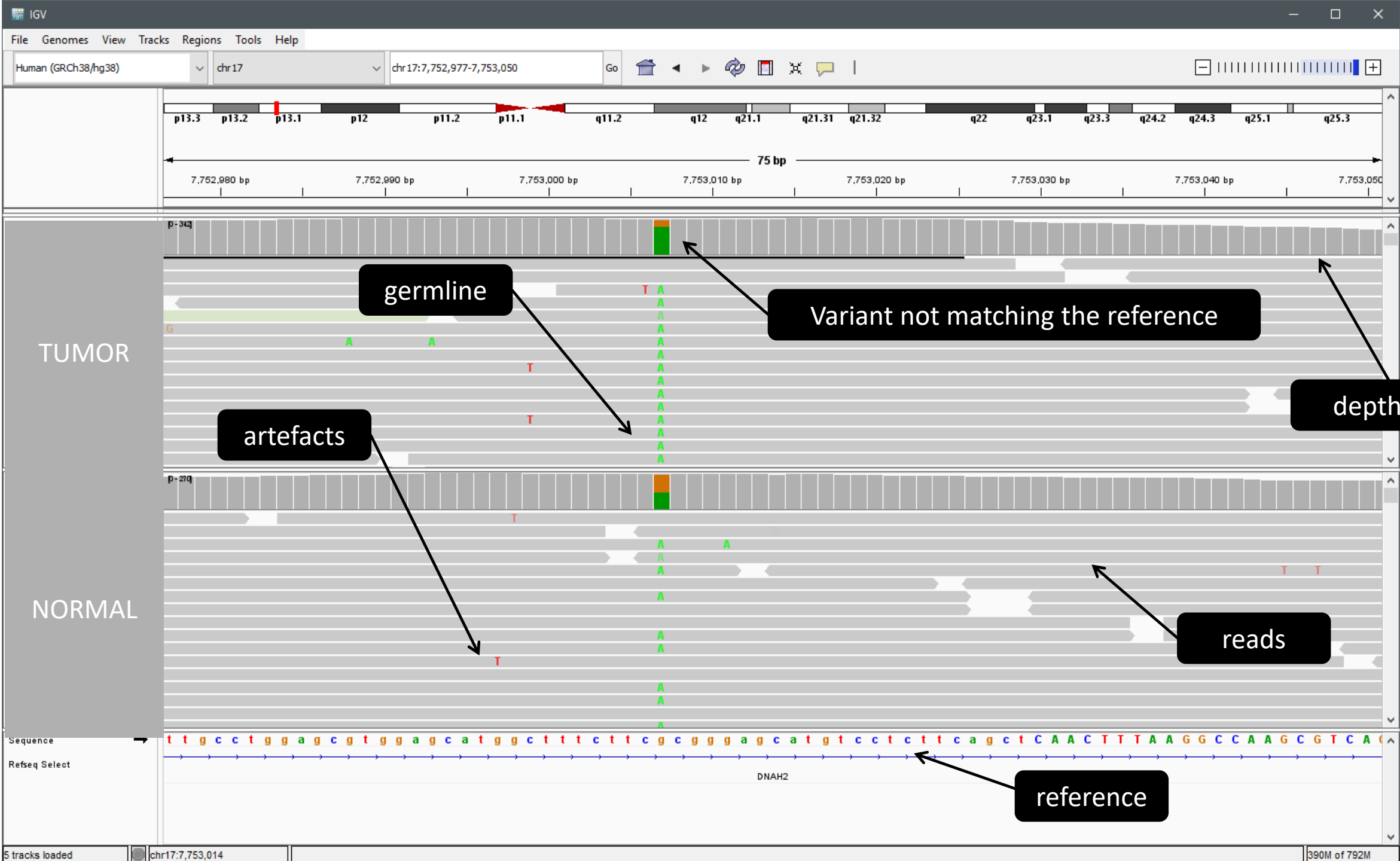
Sequencing both tumor and matched normal samples is best for the detection of somatic mutations

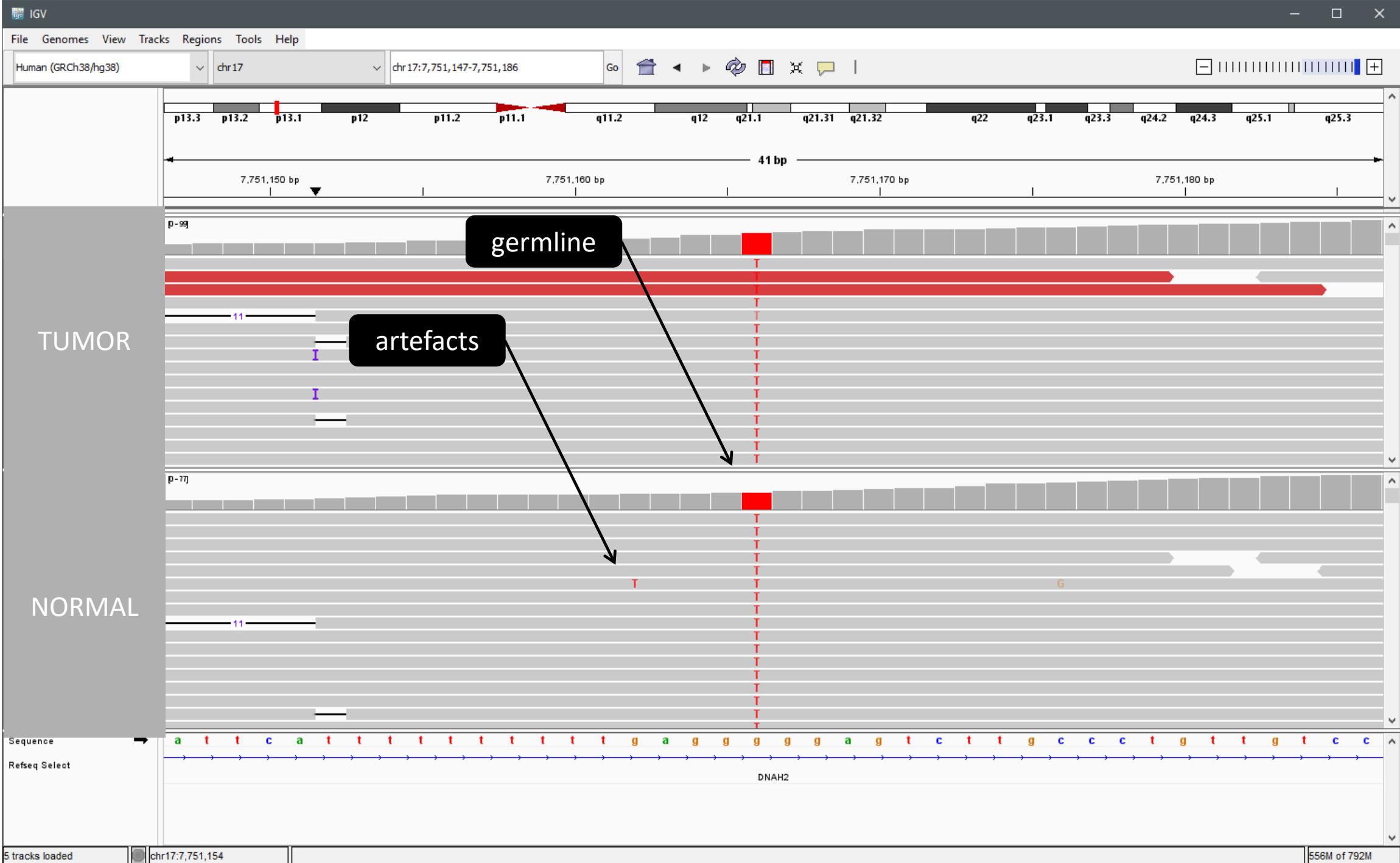


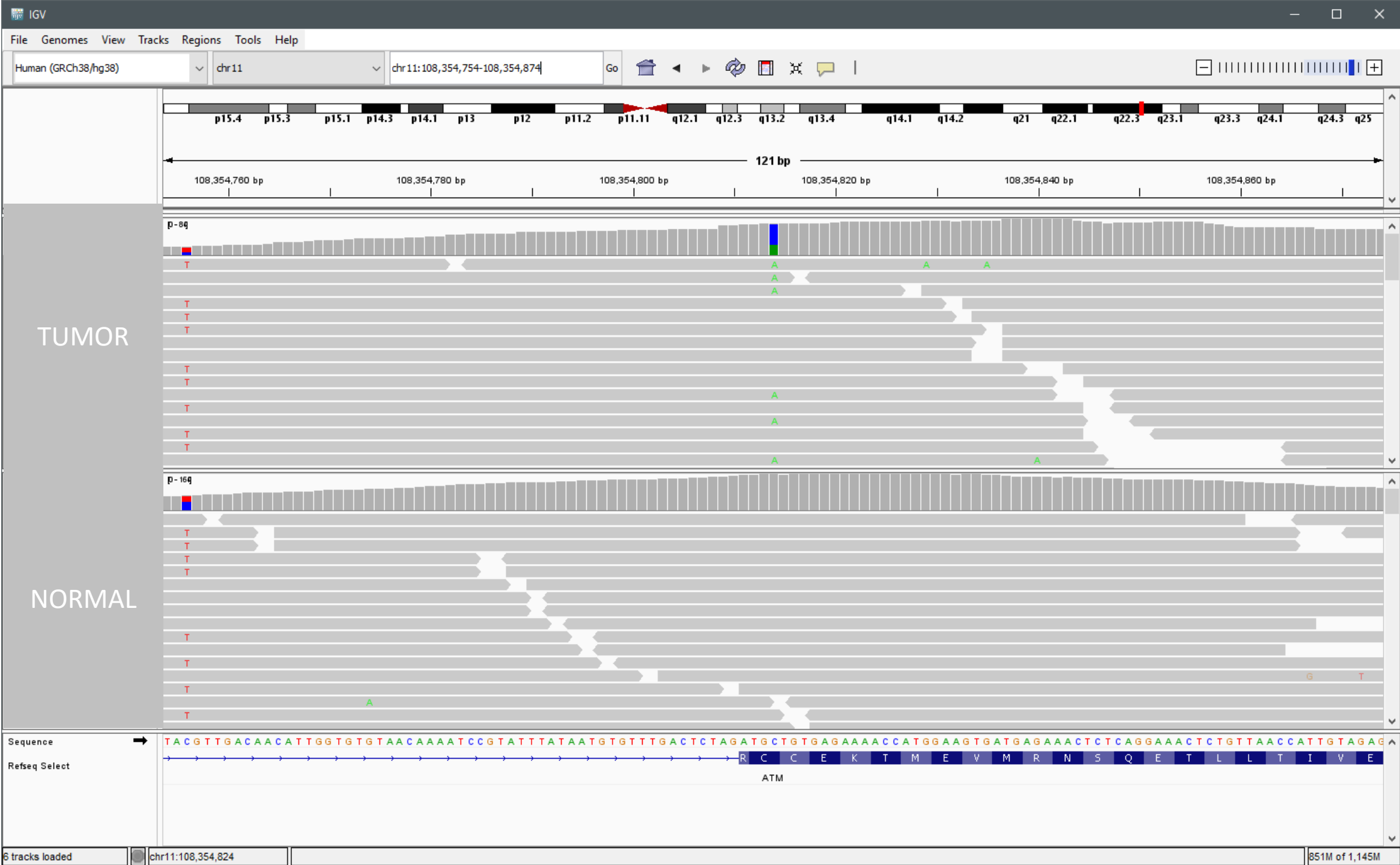
tumor and matched normal sample → somatic variant caller jointly tests T and N

tumor only 

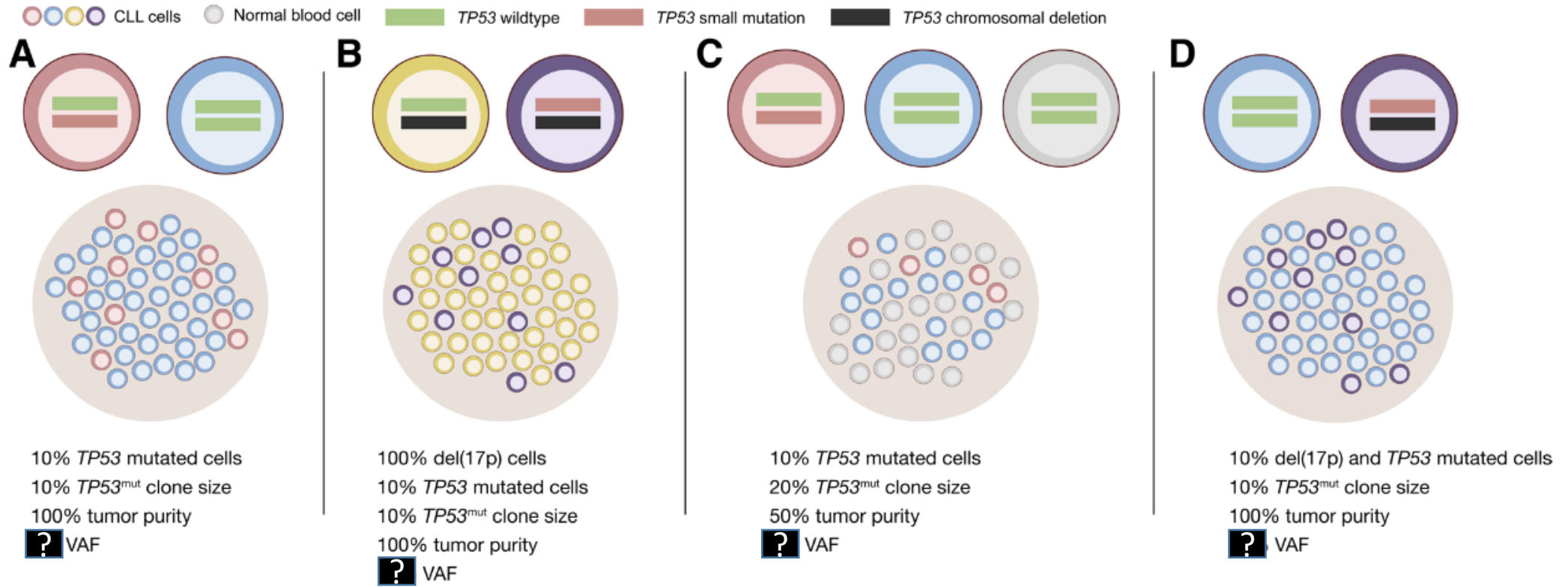
- somatic variant caller jointly tests T and (pool of) unmatched N
- single sample somatic variant caller test T alone







Variable VAF in tumor



One cannot simply look at a **VAF** and know the **clone size** or vice-versa. You must account for **tumor purity** and the **copy number** of the gene to accurately interpret your results.

VCF File Structure

```
##fileformat=VCFv4.1
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```

Example

VCF header

Body

```
##fileformat=VCFv4.0
##fileDate=20100707
##source=VCFtools
##reference=NCBI36
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality (phred score)">
##FORMAT=<ID=GL,Number=3,Type=Float,Description="Likelihoods for RR,RA,AA genotypes (R=ref,A=alt)">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##ALT=<ID=DEL,Description="Deletion">
##INFO=<ID=SVTYPE,Number=1,Type=String,Description="Type of structural variant">
##INFO=<ID=END,Number=1,Type=Integer,Description="End position of the variant">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT SAMPLE1 SAMPLE2
1 1 . ACG A,AT . PASS . GT:DP 1/2:13 0/0:29
1 2 rs1 C T,CT . PASS H2;AA=T GT:GQ 0|1:100 2/2:70
1 5 . A G . PASS . GT:GQ 1|0:77 1/1:95
1 100 T <DEL> . PASS SVTYPE=DEL;END=300 GT:GQ:DP 1/1:12:3 0/0:20
```

Mandatory header lines

Optional header lines (meta-data about the annotations in the VCF body)

Reference alleles (GT=0)

Alternate alleles (GT>0 is an index to the ALT column)

Deletion

SNP

Large SV

Insertion

Other event

Phased data (G and C above are on the same chromosome)

Genotype

```
FORMAT      NA00001
GT:GQ:DP:HQ 0|0:48:1:51,51
GT:GQ:DP:HQ 0|0:49:3:58,50
GT:GQ:DP:HQ 1|2:21:6:23,27
GT:GQ:DP:HQ 0|0:54:7:56,60
GT:GQ:DP     0/1:35:4
```

vcf

Homozygous Reference (HOM-REF)

Definition: *The individual has two copies of the reference allele at a particular genetic locus.*

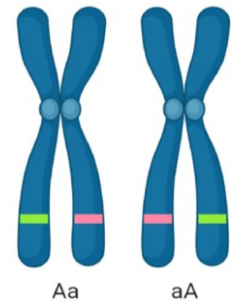
Meaning: *The individual's genotype matches the reference genome at that position, and there is no known variation.*



Heterozygous (HET)

Definition: *The individual has two different alleles at a particular locus.*

Meaning: *One allele is the reference allele, and the other is an alternative allele.*



Homozygous Alternate (HOM-ALT)

Definition: *The individual has two copies of the alternative allele at a particular genetic locus.*

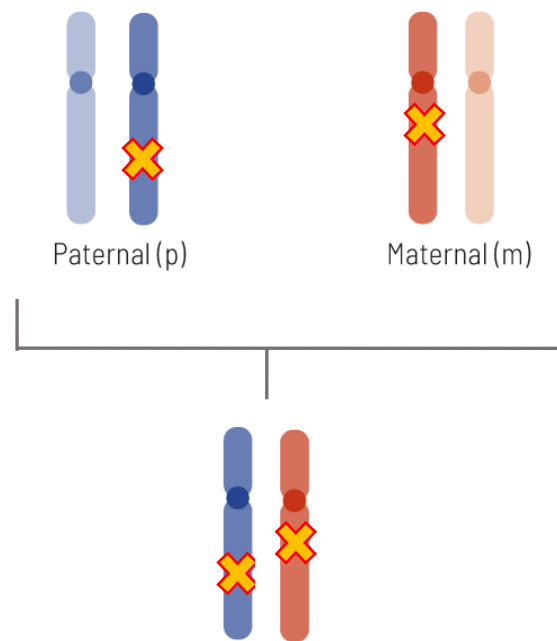
Meaning: *Both alleles are different from the reference, representing a variant that is present in both copies.*



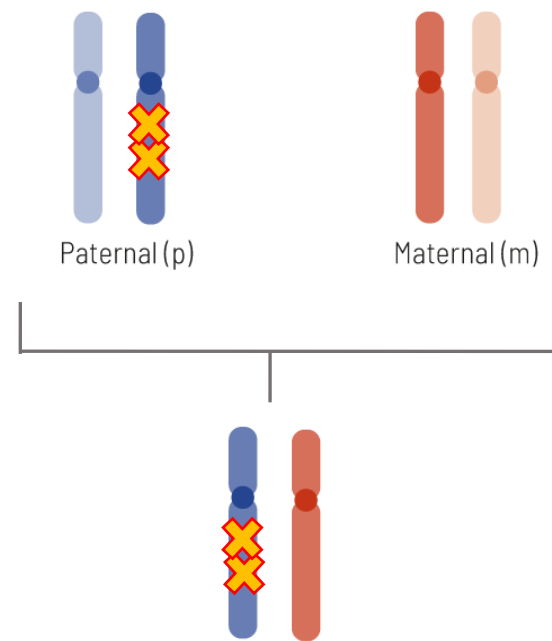
Phased variants

```
##fileformat=VCFv4.1
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT NA00001 NA00002 NA00003
20 14370 rs6054257 G A 29 PASS NS=3;DP=14;AF=0.5;DB;H2 GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20 17330 . T A 3 q10 NS=3;DP=11;AF=0.017 GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3 0/0:41:3
20 1110696 rs6040355 A G,T 67 PASS NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2 2/2:35:4
20 1230237 . T . 47 PASS NS=3;DP=13;AA=T GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20 1234567 microsat1 GTC G,GTCT 50 PASS NS=3;DP=9;AA=G GT:GQ:DP 0/1:35:4 0/2:17:2 1/1:40:3
```


Phasing

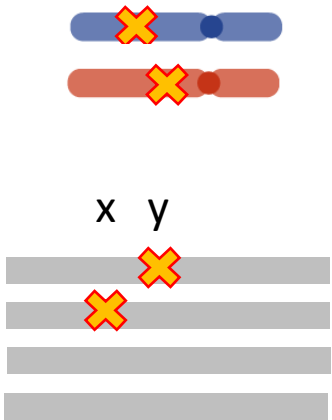


both copies affected
disease

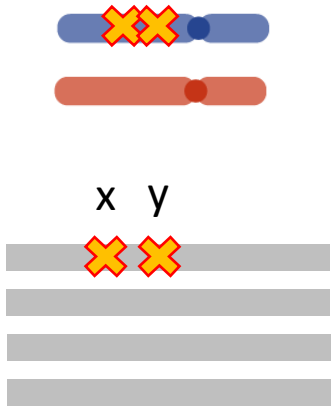
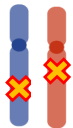


only one copy affected
No disease

Phasing



CHROM	POS	ID	S1
chr20	101	snp_x	0 1
chr20	204	snp_y	1 0



CHROM	POS	ID	S1
chr20	101	snp_x	1 0
chr20	204	snp_y	1 0



Variant calling & Annotation

Part II

16.10.2025

Institute of Tissue Medicine and Pathology – University of Bern
Clinical Genomics Lab - **INSELSPITAL**

Github Cancer Genomics Course



<https://github.com/supermax1234/CancerGenomicsCourse>

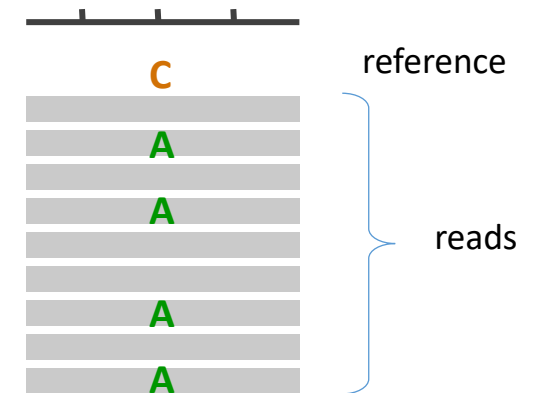
Genotype calculation (germline)

The Model: Binomial Distribution

The formula for the binomial probability is:

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k} = \frac{n!}{k!(n - k)!} \cdot p^k \cdot (1 - p)^{n-k}$$

- **Number of trials** (n): The total number of reads sequenced at this position. Here, $n = 9 = 5(ref) + 4(alt)$
- **Number of successes** (k): The number of times we observe the alternate allele. Here, $k=4$.
- **Probability of success** (p): The theoretical probability of observing the alternate allele, given the model. For a diploid, heterozygous individual, we have one reference allele and one alternate allele, so the theoretical probability is $p=0.5$.



What is the chance of seeing exactly 4 alternate reads out of 9 total reads, **if the true genotype is heterozygous**?

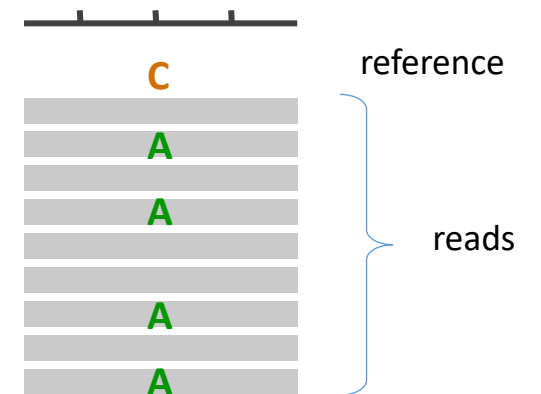
$$P(X = 4) = \binom{9}{4} \cdot 0.5^4 \cdot (1 - 0.5)^{9-4} = 126 \cdot 0.5^4 \cdot 0.5^5 \approx 0.246$$

This is the probability of the observed data given the assumed model.

The probability of observing 4 alternate reads out of 9, assuming a true heterozygous variant, is approximately 24.6%.

The probability of observing 4 alternate reads out of 9, assuming a hom. ref?

The probability of observing 4 alternate reads out of 9, assuming a hom. alt?



Likelihood calculation

Likelihood is the probability of the model given the observed data.

The numerical value/formula is the same as the probability we just calculated, but the interpretation is different.

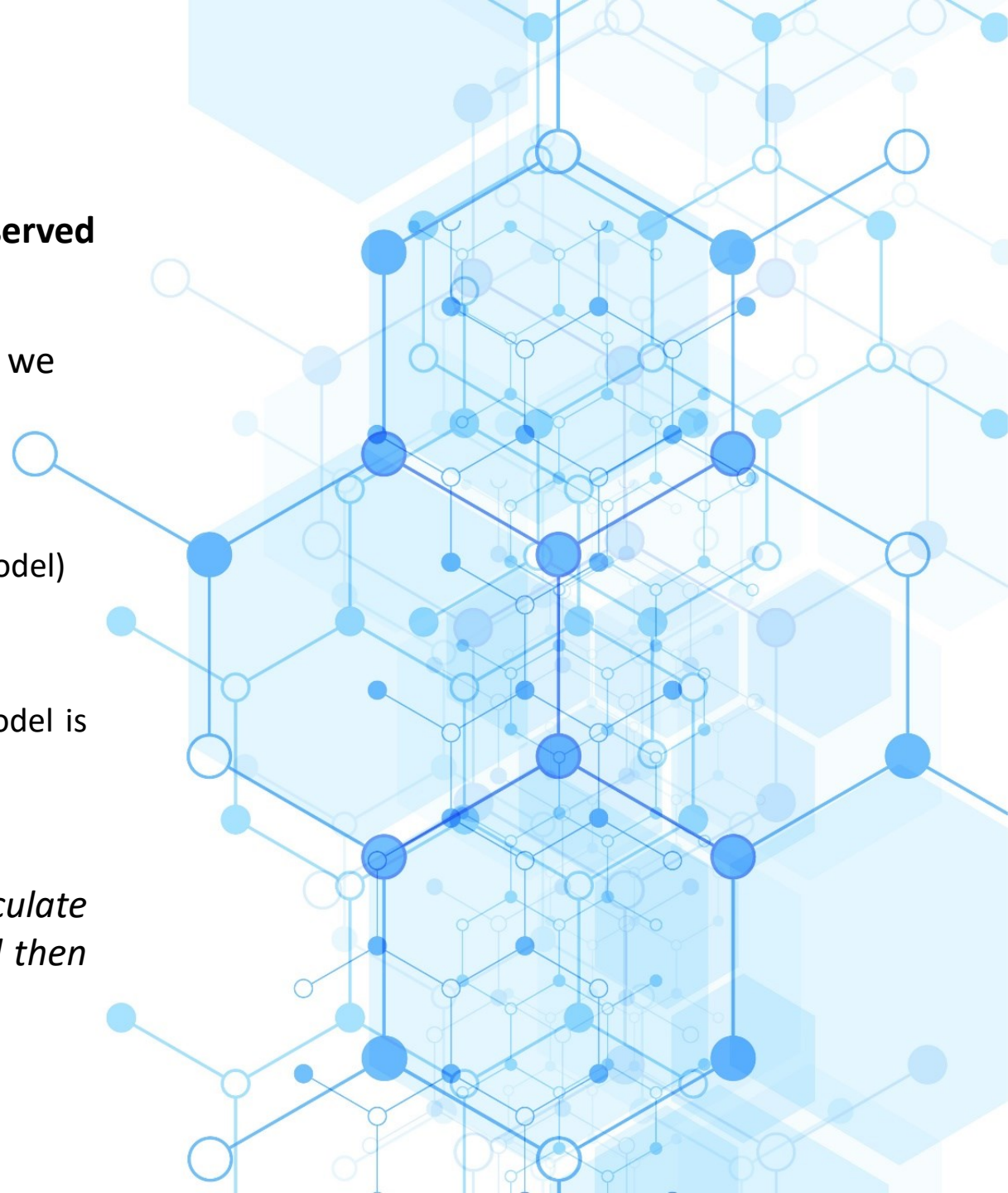
Probability answers:

What is the chance of this data occurring? (given a known model)

Likelihood answers:

How well does this model explain the data? (when the model is unknown)

*Variant callers don't just calculate one likelihood; they calculate the likelihood for **multiple possible genotypes** (models) and then pick the one with the **highest value**.*



Applying the Formula to Variant Calling

In your example, the model parameter θ is the true allele frequency (p) of the variant. The observed data x is the number of alternate reads (k) out of the total number of reads (n). So, the likelihood formula for a variant call is the same as the binomial probability formula:

$$L(p|k, n) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

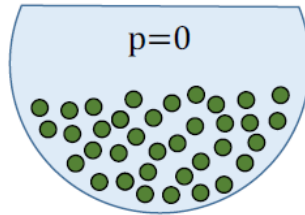
The formula for likelihood is the same mathematical function as the probability formula, but the interpretation and what you are solving for are different. The general formula for the likelihood of a parameter θ , given some observed data x , is:

$$L(\theta|x) = f(x|\theta)$$

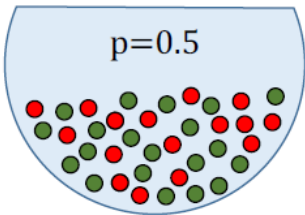


probability mass function (or density function) that you use to calculate the probability of the data. For variant calling, this is the binomial probability formula.

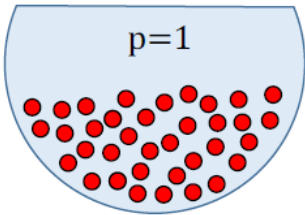
likelihood function. It represents how plausible a specific model parameter θ is, given your observed data x .



Likelihood (Homozygous Reference): What is the chance of observing 4 alternate reads if the true genotype is reference/reference? ($p=\epsilon$) -> Likelihood is $\sim 1.2\text{E-}6$.



Likelihood (Heterozygous): What is the chance of observing 4 alternate reads if the true genotype is heterozygous? ($p = 0.5$) -> Likelihood is approximately 0.246.



Likelihood (Homozygous Alternate): What is the chance of observing 4 alternate reads if the true genotype is alternate/alternate? ($p=1.0-\epsilon$) -> Likelihood is $\sim 1.2\text{E-}8$.

In this case, the **likelihood** of the **heterozygous model** being the correct one, given the data, is 0.246. Since this is the highest likelihood among the common genotypes, the variant caller would likely conclude that the genotype is indeed heterozygous.

Estimating the Genotype

This formula calculates the likelihood of a given genotype based on the *observed reads*.

It adds a crucial **layer of complexity** by explicitly modeling the sequencing error rate (ϵ).

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

where

- $\mathcal{L}(g)$ *likelihood of the assumed genotype g*
- m *ploidy. For a human genome, this is almost always $m=2$ (diploid)*
- k *total number of reads at this position*
- l *number of reads with the reference allele*
- $k - l$ *number of reads with the alternate allele*
- ϵ *sequencing error rate*
- g *count of ref alleles ($g=2 \rightarrow \text{Ref/Ref}$)*

The Logic of the Formula

The formula is a product of probabilities, meaning it calculates the likelihood by multiplying the individual probabilities of each observed read. It's essentially saying: "What is the probability of observing this exact set of k reads, given my assumption about the true genotype (g)?". The formula can be broken down into three main parts:

$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

The diagram illustrates the components of the formula. A blue arrow points from the normalization constant $\frac{1}{m^k}$ to a text box on the left. A blue bracket underneath the first product term $\prod_{j=1}^l$ is labeled "product for reference reads". Another blue bracket underneath the second product term $\prod_{j=l+1}^k$ is labeled "product for alternative reads".

This term is a normalization constant that is often dropped in practice because it is constant for a given position and doesn't change the relative likelihoods of different genotypes. It ensures the sum of all probabilities equals one, but it's not the core of the biological calculation.

- For each reference read: The formula asks, "What's the probability of this being a reference read?" It could be a correct read from the reference allele (e.g. *99% chance*), or a sequencing error on the alternate allele (e.g. *1% chance*). The formula combines these.
- For each alternate read: The formula asks, "What's the probability of this being an alternate read?" It could be a correct read from the alternate allele (e.g. *99% chance*), or a sequencing error on the reference allele (e.g. *1% chance*).

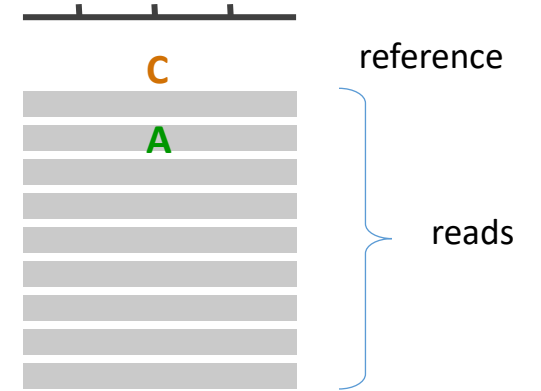
Genotype Likelihood Calculations

Total reads: $k=9$

Ref reads: $l=8$

Alt reads: $k-l=1$

Sequencing error rate: $eps=0.01$



$$\mathcal{L}(g) = \frac{1}{m^k} \prod_{j=1}^l [(m - g)\epsilon_j + g(1 - \epsilon_j)] \prod_{j=l+1}^k [(m - g)(1 - \epsilon_j) + g\epsilon_j]$$

must be a sequencing error

$$\mathcal{L}(2) = \frac{1}{2^9} [2 \cdot 0.99]^8 \cdot [2 \cdot 0.01]$$

$$\mathcal{L}(1) = \frac{1}{2^9} [0.01 + 0.99]^8 \cdot [0.99 + 0.01]$$

$$\mathcal{L}(0) = \frac{1}{2^9} [2 \cdot 0.01]^8 \cdot [2 \cdot 0.99]$$

must be a sequencing error

Phred-scaled Likelihood

$$PL = -10 \log_{10} (\mathcal{L}(g))$$

Genotype Likelihood Calculations

Total reads: $k=9$

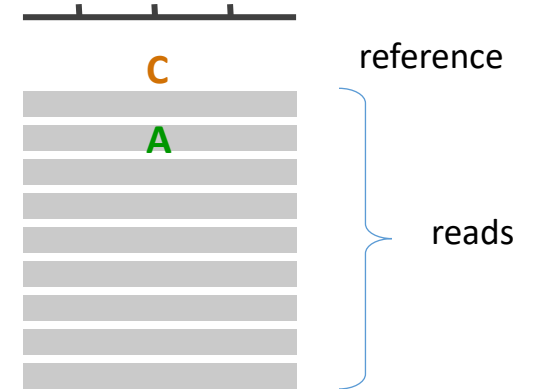
Ref reads: $l=8$

Alt reads: $k-l=1$

Sequencing error rate: $eps=0.01$

- **Likelihood (Homozygous Ref, $g=2$) = $9.2274e-03$**
PL Score = 20.35
- Likelihood (Heterozygous, $g=1$) = $1.9531e-03$
PL Score = 27.09
- Likelihood (Homozygous Alt) = $9.9000e-17$
PL Score = 160.04

Select highest Likelihood (lowest PL score) -> **Homozygous Ref**



Binomial test for allele imbalance

Is Our Heterozygote Site Truly 50:50?

The Null Hypothesis (H_0): **Allele Balance.** We assume the site is truly Heterozygous (RA), and that the probability of observing an Alt read is $p = 0.5$.

The Alternative (H_a): **Allele Imbalance.** The true proportion is $p \neq 0.5$.

The Goal: Use the binomial test to determine if the **observed read ratio** (*e.g.*, 70 Alt : 30 Ref) is statistically compatible with the $p = 0.5$ assumption, or if the imbalance is too extreme to be random chance.

Why We Only Test Heterozygotes (RA)

The Binomial Test for AI is almost exclusively used on sites called **Heterozygous (RA)**. It is unnecessary or irrelevant for the other two calls.

The Genotype Filter

<i>Genotype</i>	<i>Expected Ratio (p)</i>	<i>Test Needed?</i>	<i>Rationale</i>
Homozygous Reference (RR)	$p \approx 0$	No	Any Alt reads are already explained by the Sequencing Error Rate (ϵ) . The caller's main job is confirming the Genotype Likelihood is much higher for RR than RA or AA.
Homozygous Alternative (AA)	$p \approx 1$	No	Any Ref reads are explained by ϵ . The site is already interpreted as fully Alt.
Heterozygous (RA)	$p = 0.5$	YES	This is the ambiguous state . If the observed ratio strongly deviates from 0.5 (e.g., 70:30), it indicates a phenomenon that is not random sequencing noise.

For RR and AA, we test against the error model. For RA, we test against the 50:50 biological model.

The Binomial Logic: Testing the Extremes

When we perform the test, we ask: "**How surprising is our result?**"

The $P(X \geq k)$ Calculation

1. **Observed Data:** $N=100$ total reads. You observed $k=70$ Alt reads.
2. **Expected Mean:** Under H_0 , the expected mean is 50 Alt reads.
3. **Why $P(X \geq 70)$?**
 - We don't calculate $P(X=70)$ because the probability of *any* single, exact outcome is very small.
 - We calculate the **cumulative probability** of seeing results that are **as extreme as $k=70$, or even more extreme** (i.e., 71,72,...,100).
 - Any result *more* deviated from the mean (like 80/20) provides even stronger evidence against the 50:50 null hypothesis.

$$p\text{-value} = 2 \times P(X \geq 70) = 2 \times \sum_{x=70}^{100} \binom{100}{x} (0.5)^{100}$$

We use $\times 2$ (**two-sided test**) because an imbalance of 70:30 (too much Alt) is just as surprising as an imbalance of 30:70 (too much Ref).

Example Result and Interpretation

The calculation confirms that the observed 70:30 ratio is not random.

Example: $N=100$, $k=70$ Alt Reads

<i>Metric</i>	<i>Result</i>	<i>Interpretation</i>
Expected Mean	50 Alt Reads	The expected count under H_0 ($p=0.5$).
Observed Deviation	+20 Reads	Our observation is 20 reads above the mean.
Calculated $P(X \geq 70)$	≈ 0.0000393	The probability of seeing 70 or more Alt reads by random chance is tiny.
Final p-value (Two-Sided)	0.0000786	Extremely low!

Conclusion

Since $p \ll 0.05$, we **reject the Null Hypothesis (H_0)**.

The result is **not consistent with a true 50:50 ratio**. We have a highly significant **Allele Imbalance (AI)** flag.

Why is the Allele Imbalance Happening?

A significant AI flag requires investigation because it can be a source of noise or a key biological finding.

Possible Reasons for a 70:30 Imbalance

Category	Phenomenon	Implication (Signal or Noise)
Technical Noise	Mapping Bias: Reads containing one allele align better to the reference genome than the other allele (e.g., due to local sequence complexity).	Noise: The genotype call (RA) is likely correct, but the read ratio is skewed by technical problems. Must be filtered.
Technical Noise	PCR/Amplification Bias: One allele is preferentially copied during library preparation, leading to inflated read counts.	Noise: Requires filtering.
Biological Signal (DNA)	Loss of Heterozygosity (LOH): In a tumor sample, the cell lost the chromosome carrying the Reference allele. The true genotype is moving toward AA.	Signal: LOH is a critical finding in cancer genetics. The initial RA call may be inaccurate or represent mixed cells.

Next Step: Flag the variant, but maintain the initial RA call unless the imbalance is severe enough (e.g., 90:10) to trigger a Genotype Likelihood re-evaluation toward AA.

Variant Calling categories

- Germline Variant Callers
- Matched Somatic Variant Callers
- Somatic Variant Callers

These three categories represent **distinct statistical models** tailored for the **specific biological problem** they are trying to solve



Germline Variant Callers

Identify Inherited Variants Present in ALL Cells of the Sample

Focus	Description	Statistical Assumption	Example Tools
Sample Source	Normal tissue (blood , saliva, reference cell line).	High Purity (Near 100%). Any deviation from expected VAFs (e.g., 50:50 for a heterozygote) is treated as a potential sequencing error (ϵ) or Allele Imbalance (AI).	GATK HaplotypeCaller (Gold standard), DeepVariant (Neural network-based), FreeBayes (Bayesian methods)
Key Output	Genotype Likelihood (GL): Determines the confidence that the site is RR, RA, or AA.	The primary statistical focus is on modeling ϵ and local phasing to find the most probable diploid genotype.	
Limitation	Cannot reliably distinguish a somatic mutation from a sequencing artifact if no matched sample is used for subtraction.		

Matched Somatic Variant Callers

Accurately Subtract Normal Noise to Find Tumor-Specific Mutations

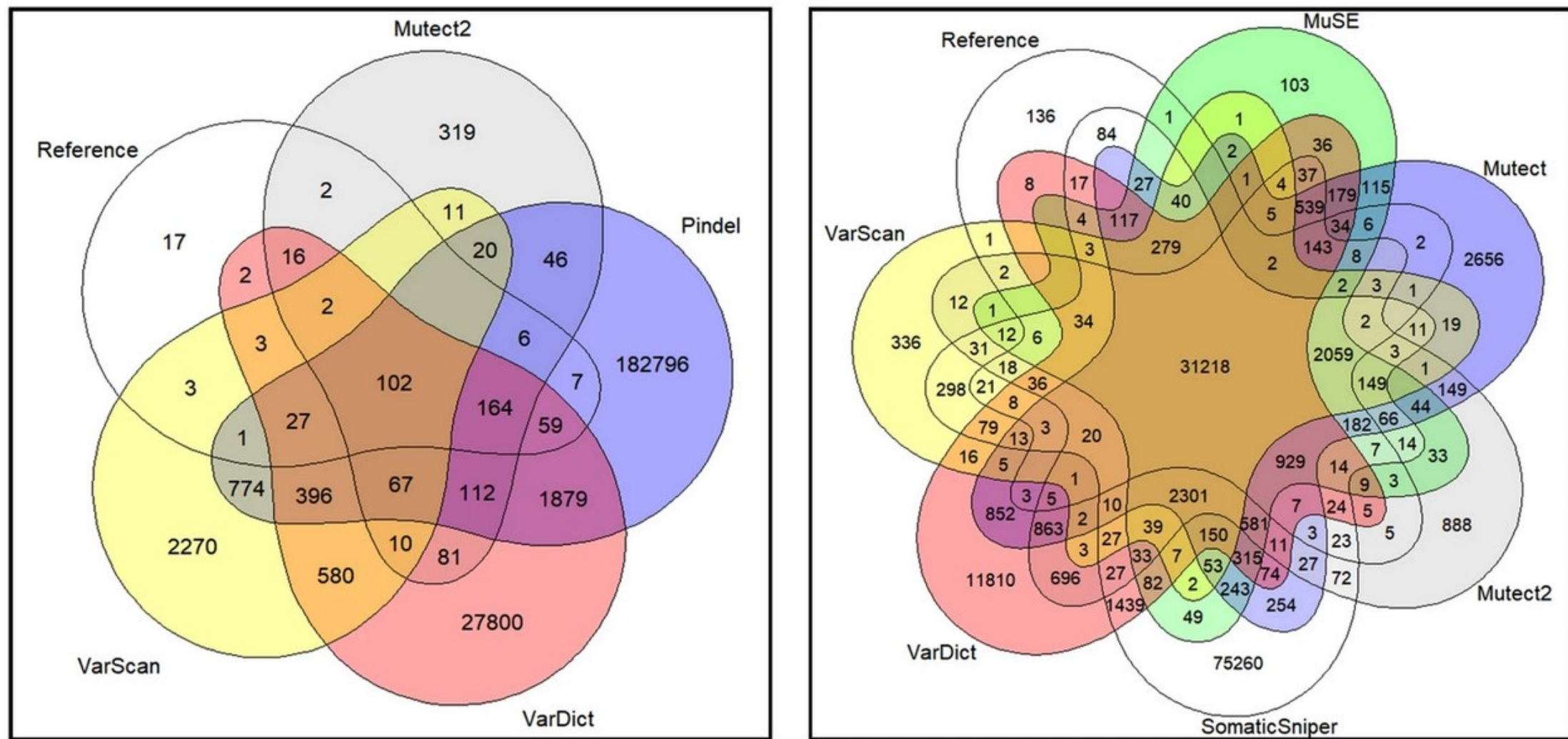
Focus	Description	Statistical Assumption	Example Tools
Sample Source	Requires two paired samples : Tumor (T) and the patient's Matched Normal (N) (e.g., blood).	The caller runs the Germline model on the Normal sample to establish the baseline of the patient's inherited genome. It then identifies variants present in T but absent in N .	GATK Mutect2 (Paired mode), Strelka2 , VarScan2 (Paired mode).
Key Advantage	Minimizes False Positives (FPs): Effectively filters out ~99% of sequencing and mapping artifacts that are common to both T and N.	The statistical power comes from comparing the Allele Fraction (VAF) in the tumor against the VAF in the matched normal (which should be 0%).	
Primary Output	Somatic VAF and Somatic Likelihood score, not a genotype. the Somatic Likelihood Score is a statistical measure of the confidence that a variant is truly tumor-specific (somatic) and not a common artifact or an inherited variant		

Somatic Variant Callers

Identify Somatic Variants When a Matched Normal is Unavailable

Focus	Description	Key Challenge	Example Tools
Sample Source	Tumor sample only. No patient-specific baseline to compare against.	High Rate of False Positives: The caller must aggressively filter out known Germline SNPs (using databases like dbSNP/gnomAD) and rely heavily on local sequencing quality to reject artifacts.	Mutect2 (Single-sample mode), VarScan2 (Single-sample mode), LoFreq .
Filtering Strategy	Database Exclusion: If a variant is in gnomAD (common population variant), it is assumed to be Germline and filtered out.	Germline variants not in public databases (rare inherited variants) will be called as somatic FPs.	
When Used	Typically reserved for cases where a matched normal sample is impossible to obtain , or for rapid screening where high sensitivity is prioritized over low false-positive rates.	Allele Imbalance is a more significant filter here than in paired mode.	

From: [Simple combination of multiple somatic variant callers to increase accuracy](https://www.nature.com/articles/s41598-023-34925-y)



Intersections between reference variants and indels (left) and SNVs (right) called by variant callers. Ellipse and overlapping area sizes are approximately proportional to variant counts.

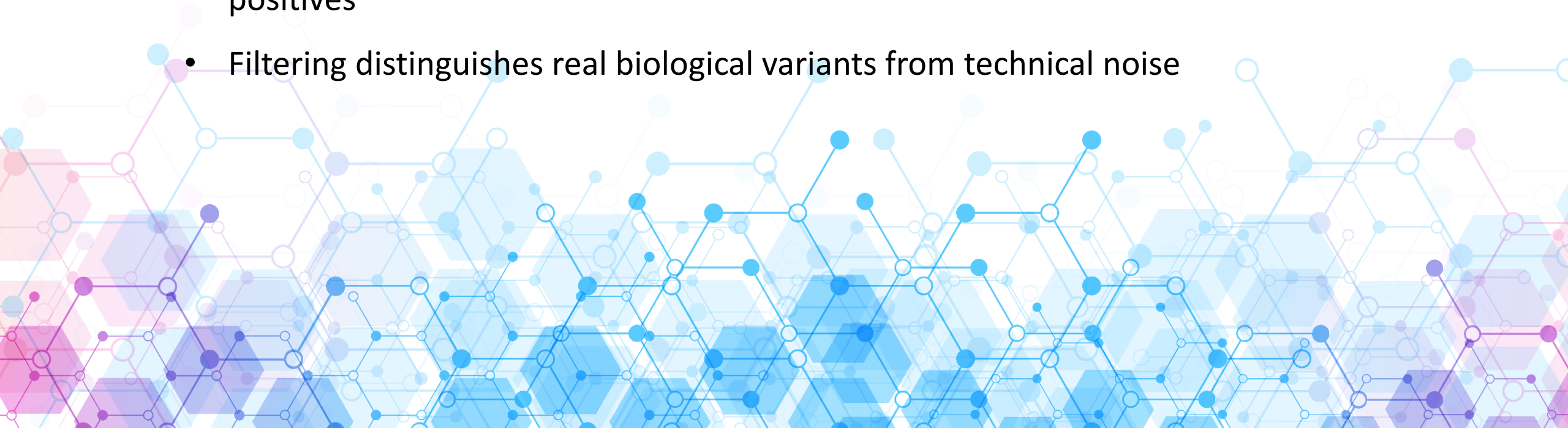


SECTION BREAK

3. Variants filtering

Why do we filter variants?

- Variant callers are **probabilistic** → they output all possible variants, not only true ones
- **Sequencing errors, mapping errors, and sample artifacts** create false positives
- Filtering distinguishes real biological variants from technical noise



Key Variant Quality Metrics (GATK VariantAnnotator)

FisherStrand	Strand bias estimated using Fisher's Exact Test. Tests if variant appears more on one strand
MappingQualityRankSumTest	Rank Sum Test for mapping qualities of REF versus ALT reads.
QualByDepth	Variant confidence normalized by unfiltered depth of variant samples
RMSMappingQuality	Root mean square of the mapping quality of reads across all samples
ReadPosRankSumTest	Rank sum test for relative positioning of REF versus ALT alleles within reads
StrandOddsRatio	Strand bias estimated by the symmetric odds ratio test
BaseQualityRankSumTest	Rank sum test of REF versus ALT base quality scores

Key Variant Quality Metrics (GATK VariantAnnotator)

AlleleFraction	Variant allele fraction for a genotype
BaseQuality	Median base quality of bases supporting each allele
CountNs	Number of Ns at the pileup
DepthPerAlleleBySample	Depth of coverage of each allele per sample
Coverage	Total depth of coverage per sample and over all samples
dbsnp	flag if it is in the dbsnp database

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	tumor	normal
chr1	69897	rs200676709	T			C	. MQ_lower_40;alt_allele_in_normal;low_normal_depth;weak_evidence	AQ=50.53;AS_BaseQRankSum=0.674;AS_MQ=28.64;AS_MQRankSum=0.000;AS_ReadPosRankSum=0.524;BaseQRankSum=-1.645;DP=5		
chr1	924338		G	A			DP_lower_6;weak_evidence	AS_BaseQRankSum=-1.645;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=0.524;BaseQRankSum=-1.645;DP=5		
chr1	946463	rs752693019	G			A	. low_af;weak_evidence	AS_BaseQRankSum=0.672;AS_MQ=60.00;AS_MQRankSum=0.401;AS_ReadPosRankSum=-0.891;BaseQRankSum=0.672;DP=5		
chr1	982103	rs1277007405				G	A . low_af;weak_evidence	AS_BaseQRankSum=0.261;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=0.662;BaseQRankSum=0.261		
chr1	999140		C	A			no_reliable_supporting_read;read_position;weak_evidence	AS_BaseQRankSum=0.000;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=-1.710;BaseQRankSum=-0.021;DP=65;FS=3.323;FractionInformativeReads=0.935;MappingQuality=60		
chr1	1040817	rs1242198480				G	A . low_af;weak_evidence	AS_BaseQRankSum=-0.401;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=0.523;BaseQRankSum=-0.401		
chr1	1044016	rs754655150	C			T	. low_af;weak_evidence	AS_BaseQRankSum=1.679;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=1.868;BaseQRankSum=1.679;DB=100;DP=576;FS=0.000;FractionInformativeReads=1.0		
chr1	1046832		T	G			low_af;weak_evidence	AS_BaseQRankSum=0.679;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=-1.710;BaseQRankSum=0.679;DP=253;FS=0.000;FractionInformativeReads=1.0		
chr1	1303216	rs779727003	G			A	. low_af	AS_BaseQRankSum=0.086;AS_MQ=60.00;AS_MQRankSum=0.000;AS_ReadPosRankSum=-0.350;BaseQRankSum=0.086;DB=100;DP=576;FS=0.000;FractionInformativeReads=1.0		
chr1	1313917	rs199864556	G			A	. low_af;weak_evidence	AS_BaseQRankSum=-0.021;AS_MQ=60.00;AS_MQRankSum=0.149;AS_ReadPosRankSum=1.678;BaseQRankSum=-0.021;DP=65;FS=3.323;FractionInformativeReads=0.935;MappingQuality=60		
chr1	1471947		T	C			clustered_events;mapping_quality;no_reliable_supporting_read;weak_evidence	AS_MQ=.;DP=65;FS=3.323;FractionInformativeReads=0.935;MappingQuality=60		
chr1	1471948		TTGC	T			. clustered_events;mapping_quality;no_reliable_supporting_read;weak_evidence	AS_MQ=.;DP=68;FS=0.000;FractionInformativeReads=1.0		

vcf file

Types of Filtering Strategies

Hard Filtering

Apply simple threshold rules on metrics

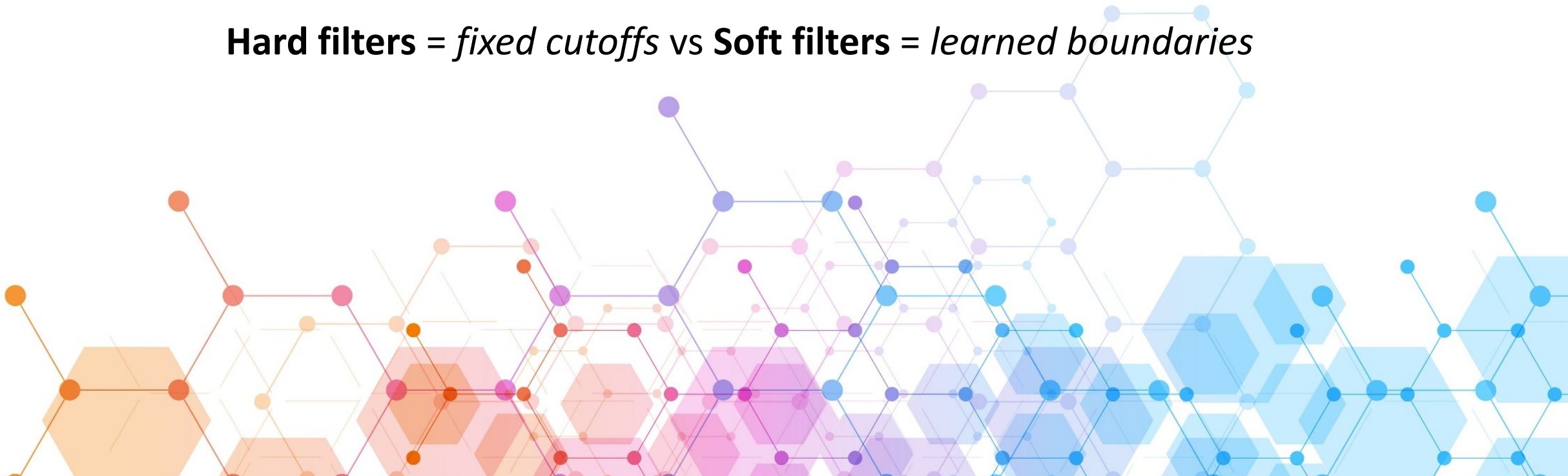
e.g. $QUAL > 30$, $DP > 10$

Soft / ML-based Filtering

Use a probabilistic model (trained from known true/false variants)

e.g. GATK VQSR, DeepVariant's internal CNN

Hard filters = *fixed cutoffs* vs **Soft filters** = *learned boundaries*



Hard Filtering : example

Example (GATK-Style Hard Filters) for SNPs

$QD < 2.0 \rightarrow \text{filter}$

$FS > 60.0 \rightarrow \text{filter}$

$MQ < 40.0 \rightarrow \text{filter}$

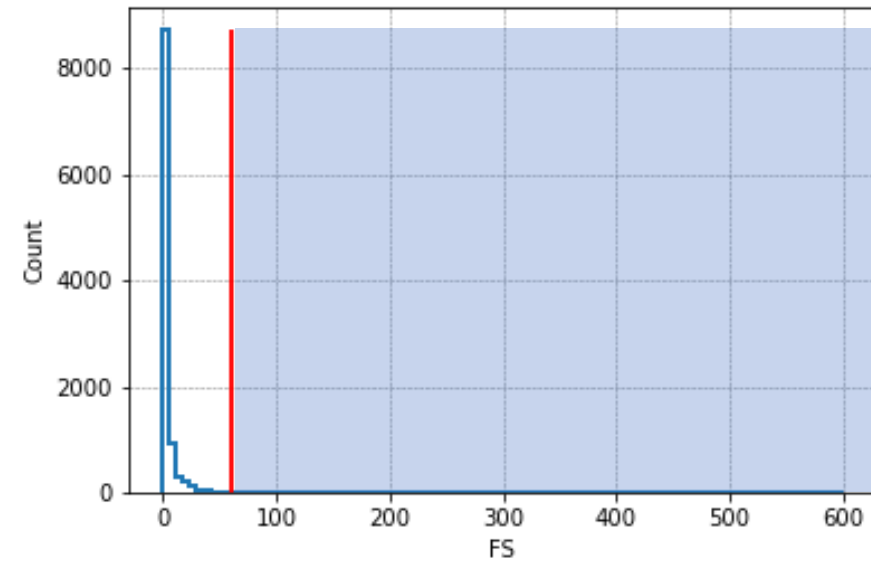
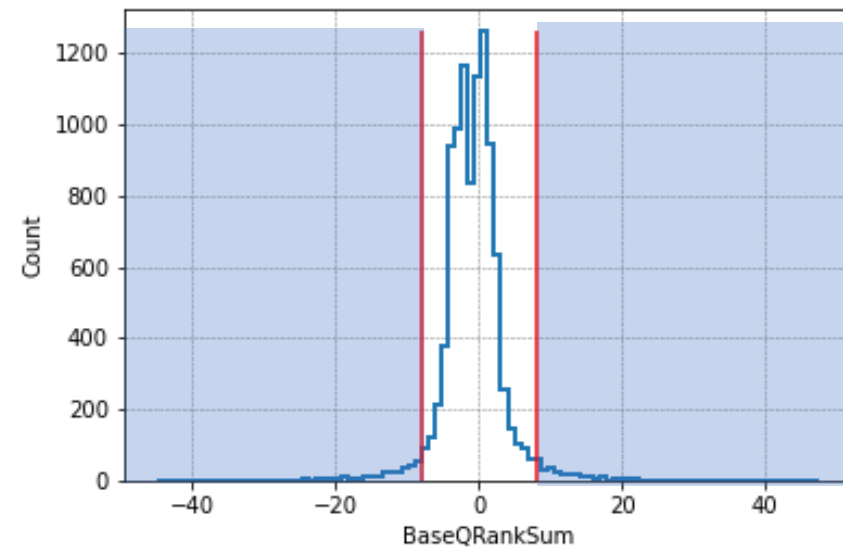
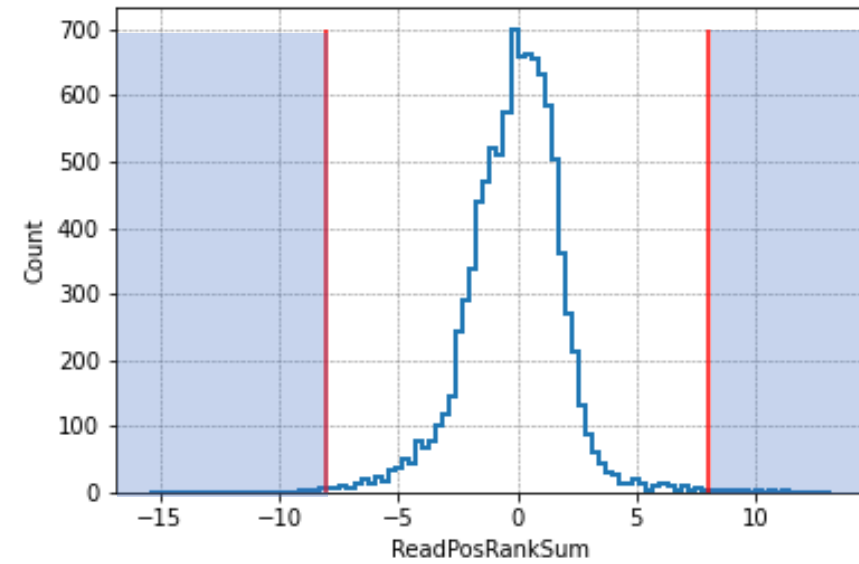
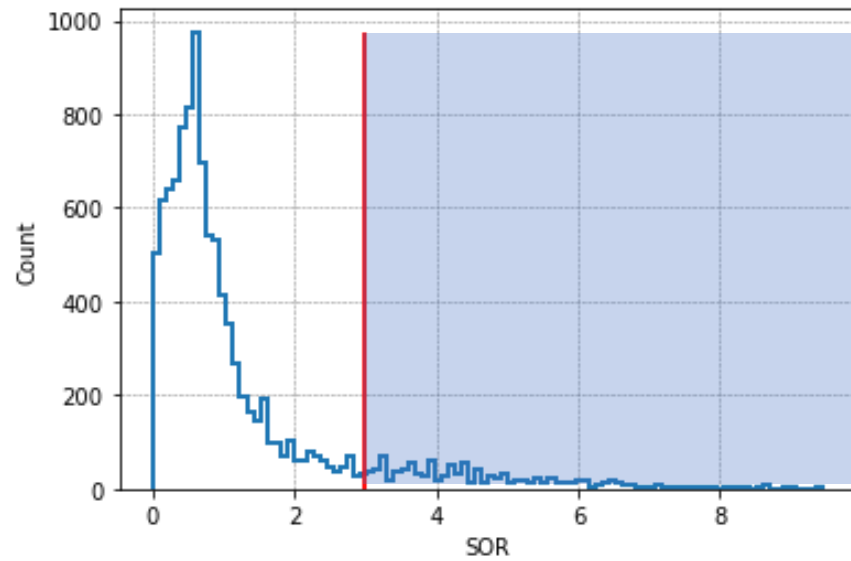
$SOR > 3.0 \rightarrow \text{filter}$

$MQRankSum < -12.5 \rightarrow \text{filter}$

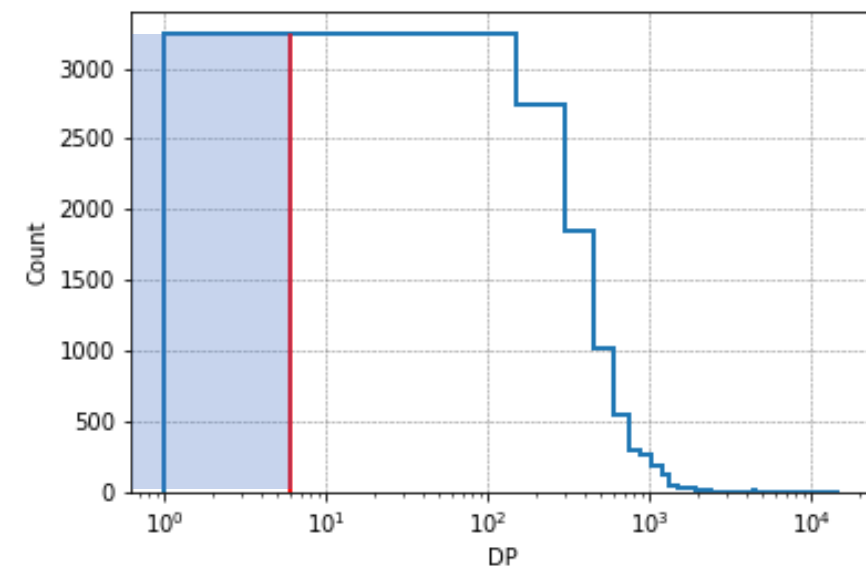
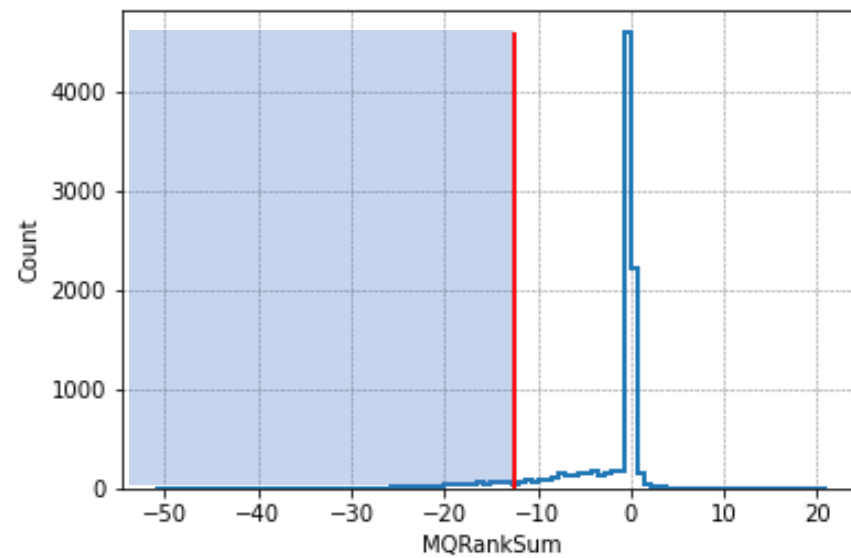
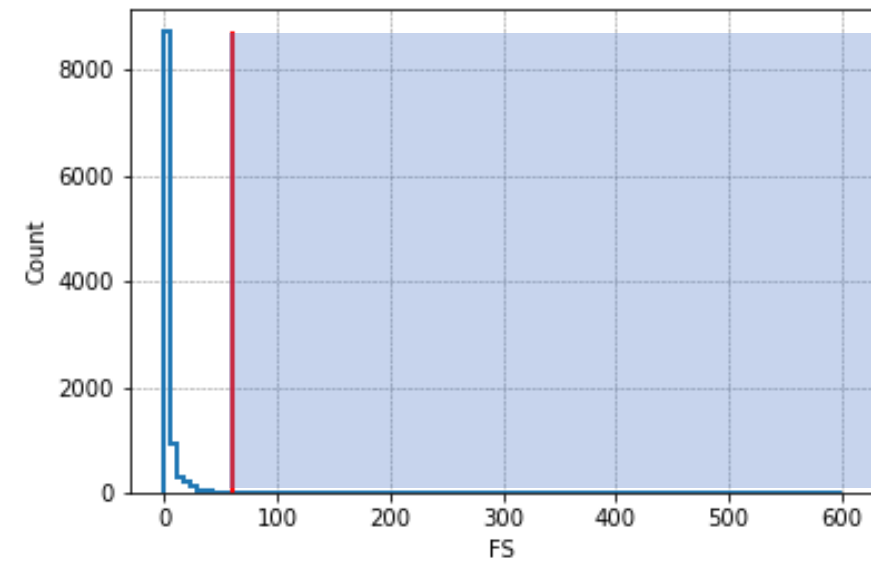
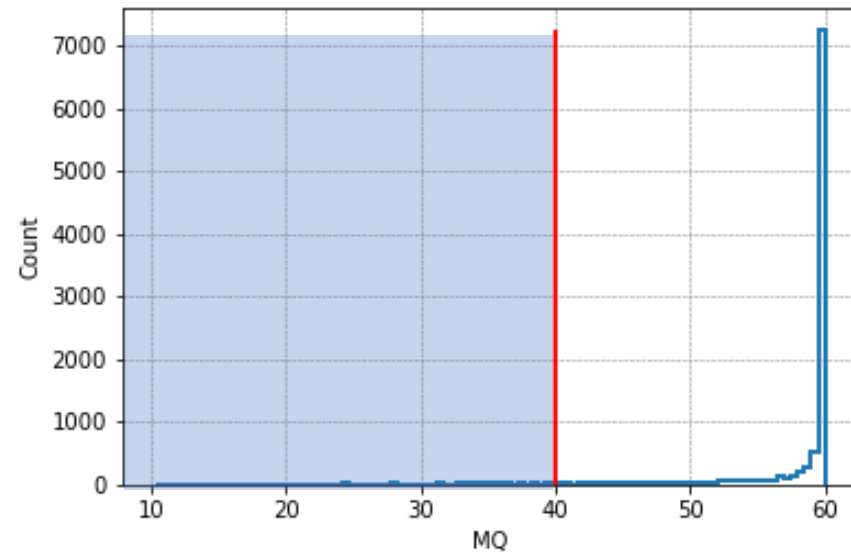
$ReadPosRankSum < -8.0 \rightarrow \text{filter}$



Rank scores distribution



Rank scores distribution



Soft Filtering — Machine Learning (VQSR)

- Uses **Gaussian Mixture Model** trained on known true variant sets (e.g. dbSNP, ...).
- Learns typical patterns of true vs false variants using metrics (QD, MQ, FS, etc.).
- Assigns a **VQSLOD score** = log-odds ratio of being true vs false.
- You then choose a **tranche** (e.g. 99.0% sensitivity) to filter variants.

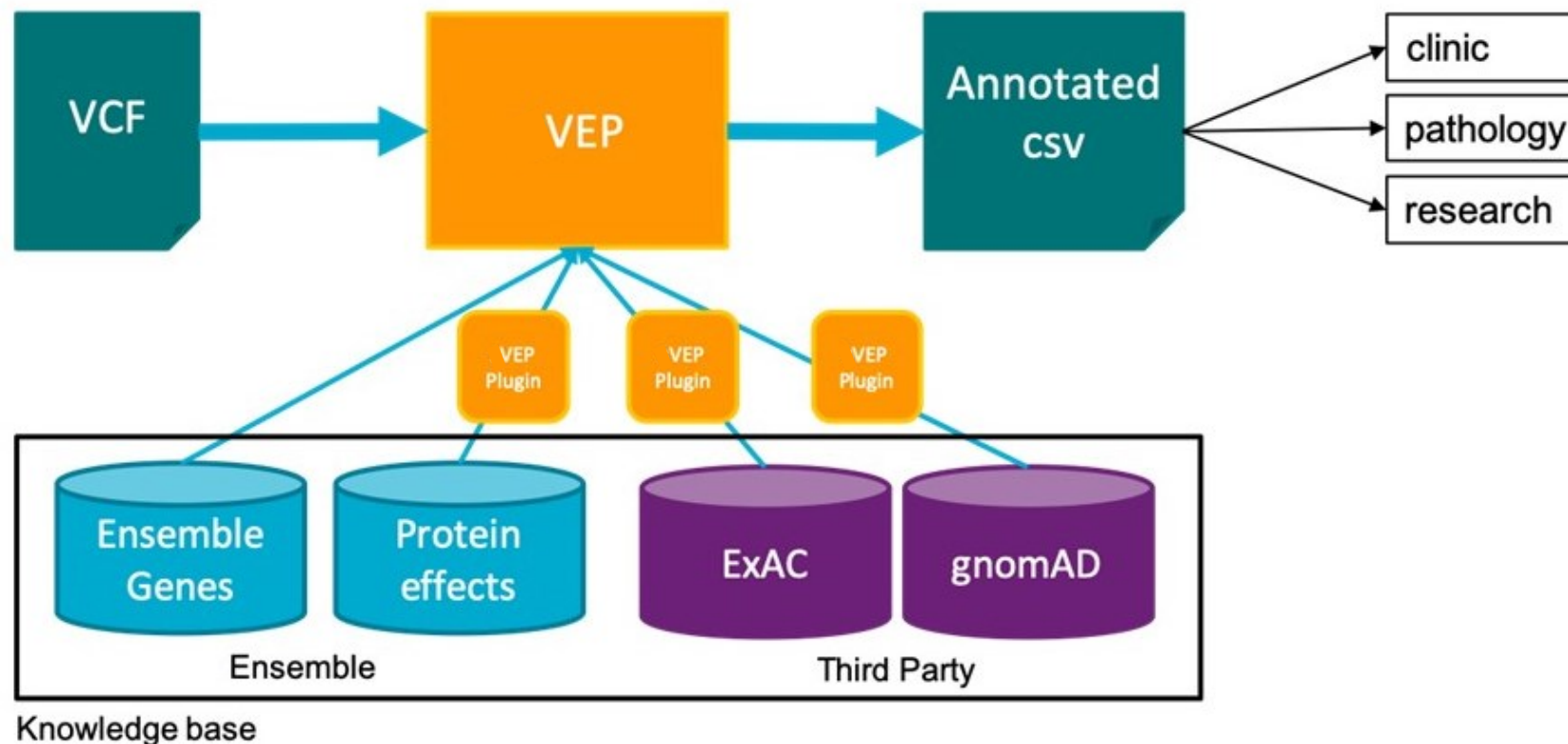
Beyond VQSR — Modern ML Filters

- **CNNScoreVariants** (GATK) – deep convolutional network using pileup images
- **DeepVariant** – neural network performs joint calling + filtering



VCF Annotation: The Bridge to Biological Filtering

VCF Annotation: Equipping Our Variants with Evidence



VCF Annotation: The Bridge to Biological Filtering

Biological / functional filtering that happens **after the technical filtering** (quality-based)

After **technical filtering** (Hard Filtering or VQSR), we have a high-confidence list of variants—meaning they are *real* and *not* sequencing errors. **However, 99% of these variants are likely benign.** Annotation is the critical step that equips each variant with the necessary external data to find the pathogenic few.

Annotation is the process of **decorating** every variant in our VCF with metadata from external databases. We are not making a call yet; we are gathering all available evidence to allow for subsequent intelligent filtering.

VEP (Variant Effect Predictor), SnpEff, ANNOVAR. These command-line tools automate the integration of dozens of public databases into new fields within the VCF's INFO column.



From Reliable Variants to Relevant Variants

This is usually called ***annotation-based or interpretative filtering*** — the step where we decide *which* variants are biologically interesting among those that are technically high-confidence.

Annotation-based filtering uses annotations from public databases to characterize each variant.

Category	Examples / Databases	Description / Purpose
Gene, transcript, consequence	Ensembl, RefSeq	Identify which gene/transcript is affected and the predicted consequence (missense, stop-gain, frameshift, etc.)
Known population frequency	gnomAD, 1000 Genomes, ExAC	Estimate how common the variant is in healthy populations (rare = more likely pathogenic)
Predicted functional impact	SIFT, PolyPhen, CADD, REVEL	In silico prediction of whether the amino acid change affects protein function or structure
Known clinical relevance	ClinVar, COSMIC, OncoKB, OMIM	Databases linking variants to diseases, cancer types, drug responses, and literature evidence

Insertion (1 base): 1-55516888-G-GA(GRCh37)

[Copy variant ID](#)[Gene page](#)Dataset [gnomAD v2.1.1](#) 

	Exomes	Genomes	Total
Filters ?	Pass	Pass	
Allele Count	121	156	277
Allele Number	128230	31382	159612
Allele Frequency	0.0009436	0.004971	0.001735
Grpmax Filtering AF ? (95% confidence)	0.01338	0.01542	
Number of homozygotes	2	0	2
Mean depth of coverage	22.1	32.1	

External Resources

- [dbSNP \(rs527413419\)](#)
- [UCSC](#)
- [ClinGen Allele Registry \(CA871474\)](#)

Feedback

[Report an issue with this variant](#)

Genetic Ancestry Group Frequencies ?

Genetic Ancestry Group	Allele Count	Allele Number	Number of Homozygotes	Allele Frequency	
▶ African/African American	251	14784	2	0.01698	
▶ Admixed American	18	25174	0	0.0007150	
▶ Remaining individuals	3	5074	0	0.0005912	
	Overall	5	62976	0	0.00007940
	North-western European	5	38446	0	0.0001301
	Bulgarian	0	248	0	0.000
	Estonian	0	4714	0	0.000
▼ European (non-Finnish)	Other non-Finnish European	0	16636	0	0.000
	Southern European	0	1398	0	0.000

NM_004333.6(BRAF):c.1799T>A (p.Val600Glu)

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We've updated the full submission template; previous versions are no longer supported. Download the latest version from [the ClinVar FTP site](#) and read more about the available templates.

Germline

Classification



Conflicting classifications of pathogenicity

Pathogenic(3); Likely pathogenic(1); Uncertain significance(1)

5 out of 21 submissions contributed to this classification



Somatic

Clinical impact



**Tier I (Strong) for 3 tumor types
+lower levels of evidence for 2 other tumor
types**

4 out of 6 submissions contributed to this classification



Somatic

Oncogenicity



Oncogenic

1 out of 2 submissions contributed to this classification

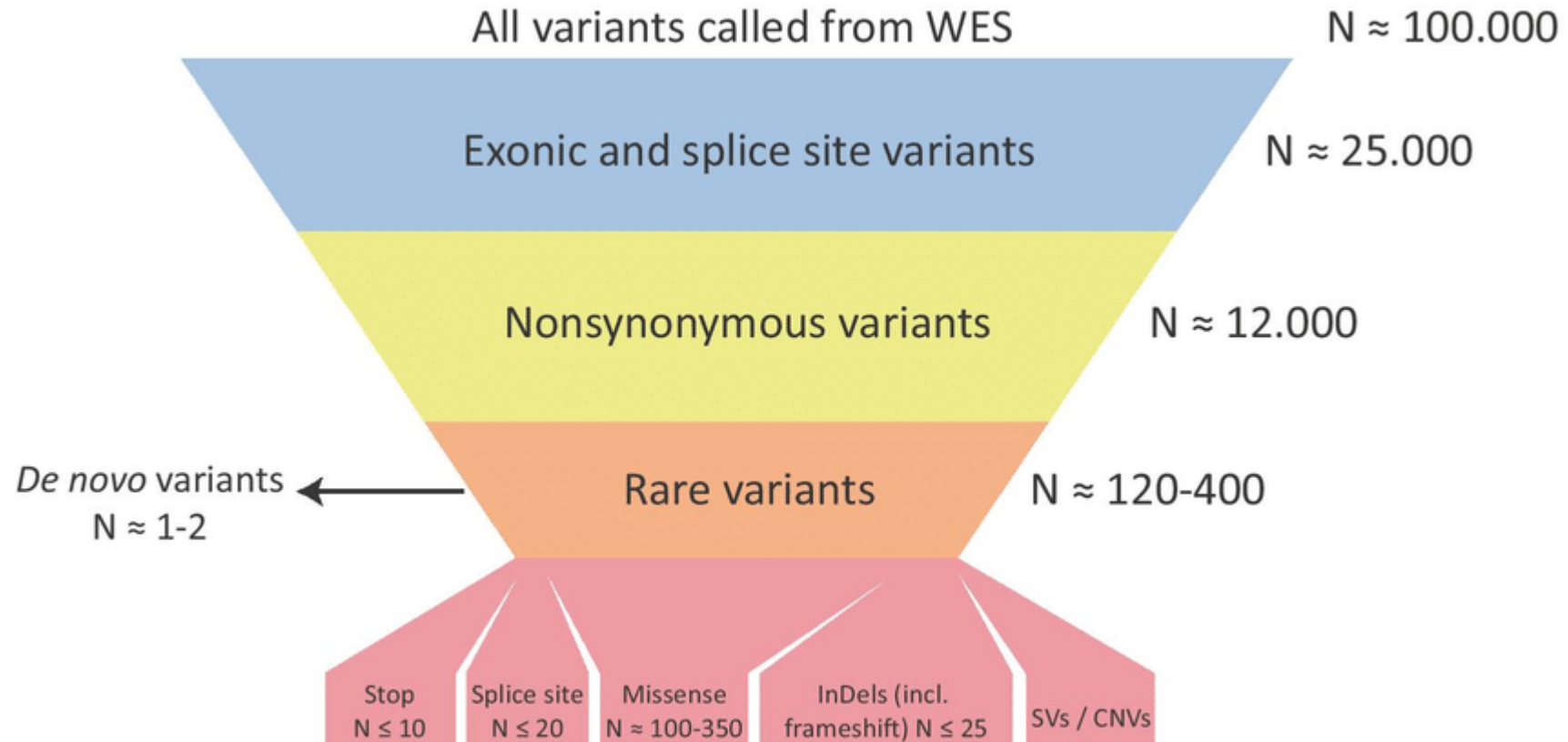


Common Functional Filters

Typical Biological Filtering Criteria

- **Genomic region:** keep only *exonic* or *splice-site* variants
- **Functional effect:** keep *non-synonymous*, *stop-gain*, *frameshift*, *splicing*
- **Population frequency:**
 - remove variants with **AF > 0.01** in gnomAD (too common to cause rare disease)
- **Clinical databases:**
 - retain variants labeled *pathogenic* or *likely pathogenic*
- **In silico predictions:**
 - keep variants predicted *deleterious* (SIFT < 0.05, PolyPhen > 0.85, CADD > 20)
- **Inheritance model filters:**
 - e.g., homozygous in recessive disorder, compound heterozygous, etc.

The general steps used to filter variants from WES



Tools for Annotation and Filtering

- **Ensembl VEP (Variant Effect Predictor)**
- **SnpEff / SnpSift**
- **ANNOVAR**
- **bcftools + filter, awk, or custom scripts**
- **Gemini, seqr, VarSome, Franklin**





END